第11章 耦合电感和理想变压器

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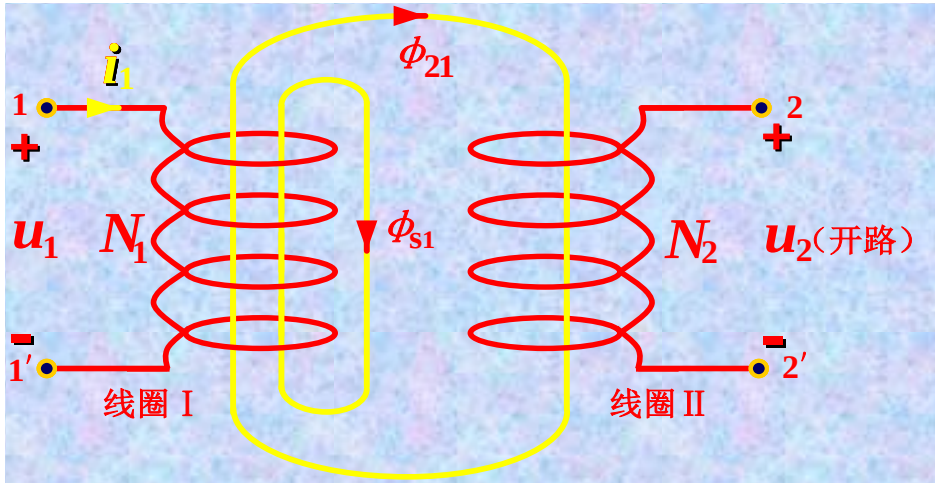
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§11-1　基本概念

耦合电感元件属于多端元件，在实际电路中，如收音机、电视机中的中周线圈、振荡线圈，整流电源里使用的变压器等都是耦合电感元件，熟悉这类多端元件的特性，掌握包含这类多端元件的电路问题的分析方法是非常必要的。

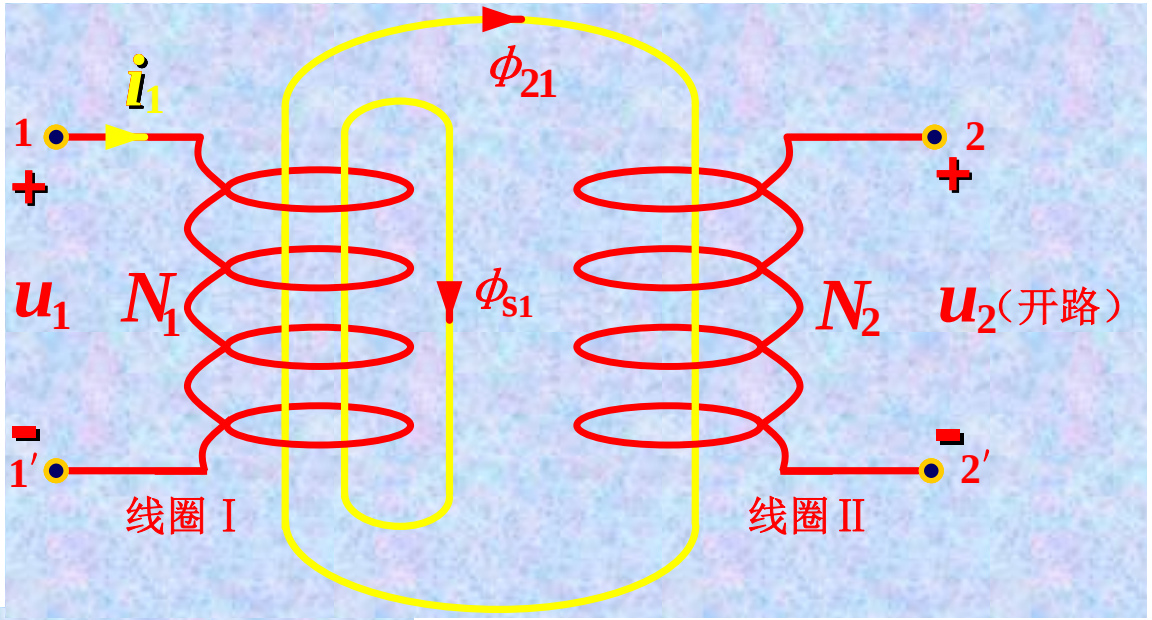


线圈 I 中通入电流i_1时，在线圈 I 中产生磁通(*magnetic flux*)，同时，有部分磁通穿过临近线圈II，这部分磁通称为互感磁通。两线圈间有磁的耦合。

§11-1 基本概念

(1) 互感电压

$$i_1 \rightarrow \Phi_{21} \rightarrow u_2$$

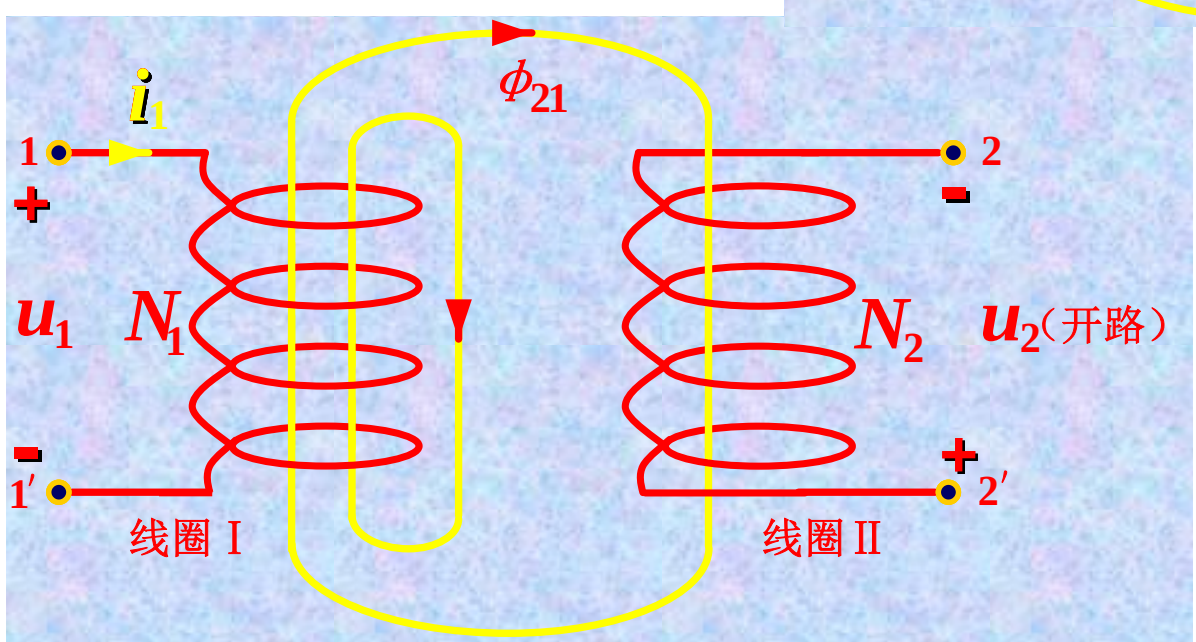


(a)

两次运用右手螺旋法则，确定i_1、u_2的参考方向后，方可运用

$$u_2 = \frac{\mathrm{d}N_2\Phi_{21}}{\mathrm{d}t} = M\frac{\mathrm{d}i_1}{\mathrm{d}t}$$

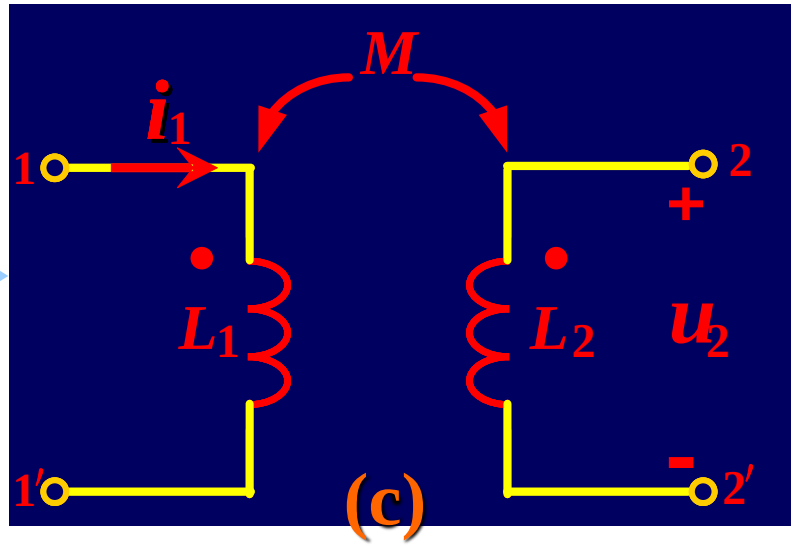
(公式1)

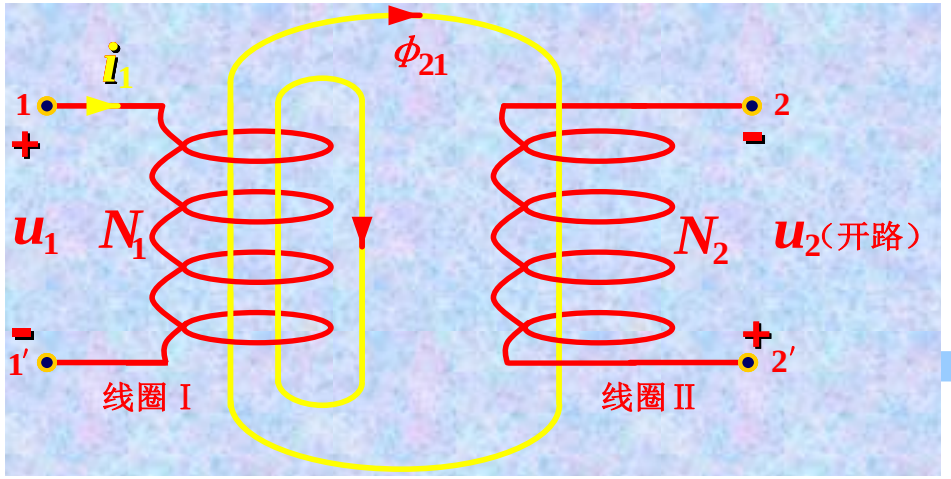


(b)

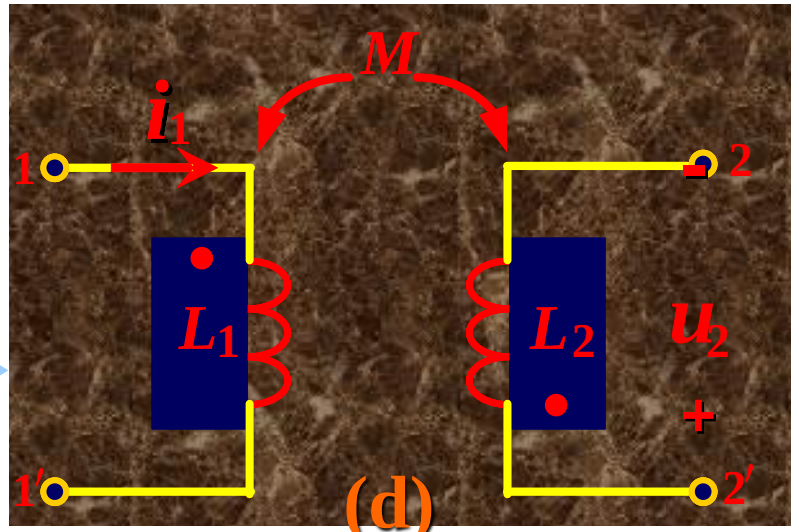
(2) 同名端

为避免如实绘出线圈绕线方向和相互位置，采用“●”的标志，以便运用公式1。





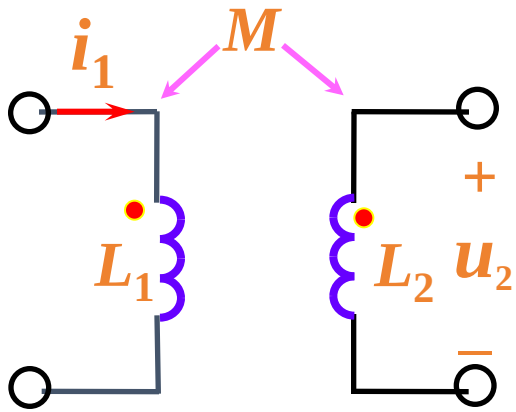
(b)



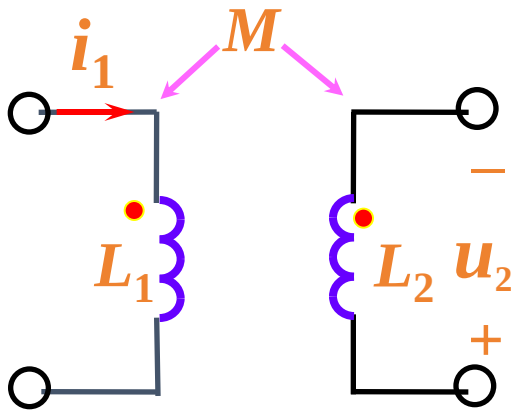
i_1流入端和u_2的正极端各标以同名端标志时方可使用公式1。

(3) 根据同名端确定互感电压的正负

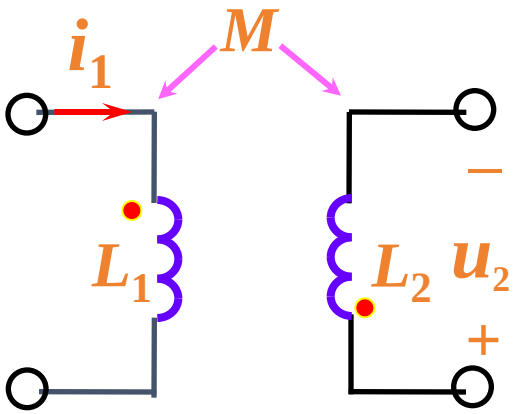
如果电流的参考方向由线圈 的同名端指向另一端，那么由这个电流在另一个线圈中产生的互感电压的 参考方向也应该由线圈的同名端指向另一端。



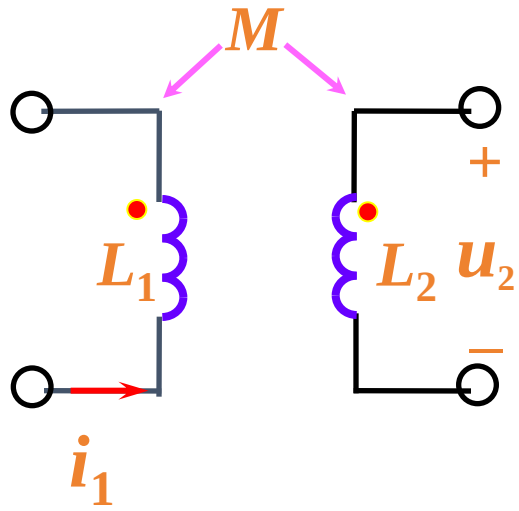
$$u_2 = M\frac{\mathrm{d}i_1}{\mathrm{d}t}$$



$$u_2 = -M\frac{\mathrm{d}i_1}{\mathrm{d}t}$$

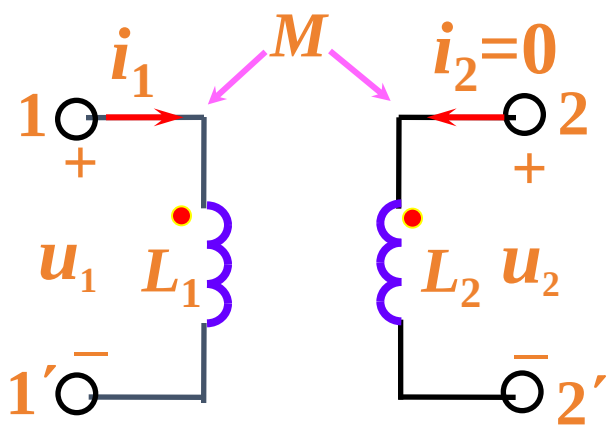


$$u_2 = M\frac{\mathrm{d}i_1}{\mathrm{d}t}$$

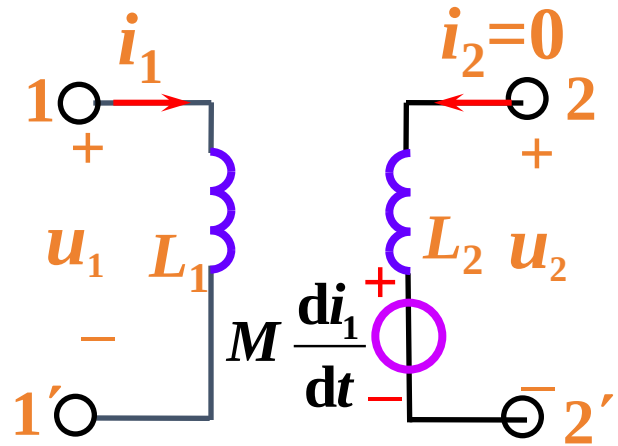


$$u_2 = -M\frac{\mathrm{d}i_1}{\mathrm{d}t}$$

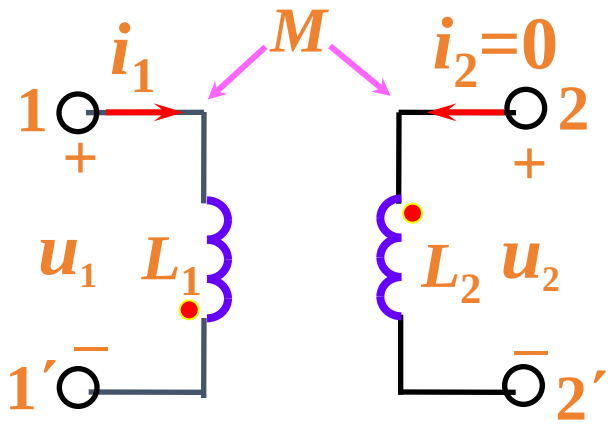
(4) 互感电压用附加的电压源代替



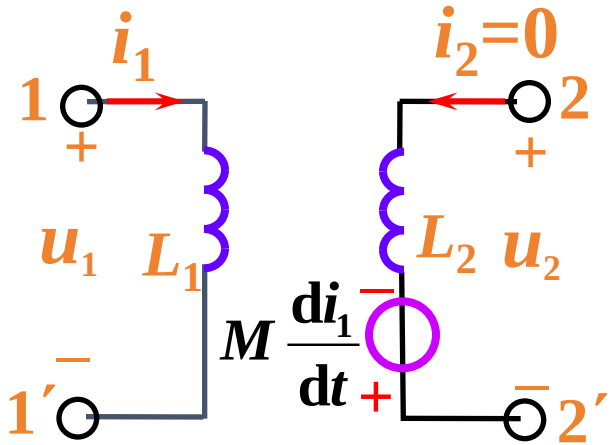
$$u_2 = M\frac{\mathrm{d}i_1}{\mathrm{d}t}$$



$$u_2 = M\frac{\mathrm{d}i_1}{\mathrm{d}t}$$

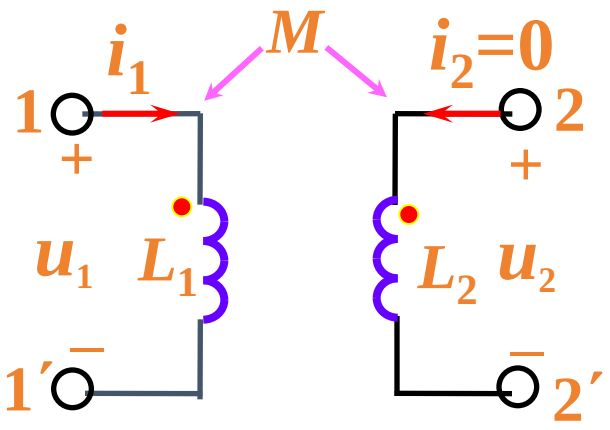


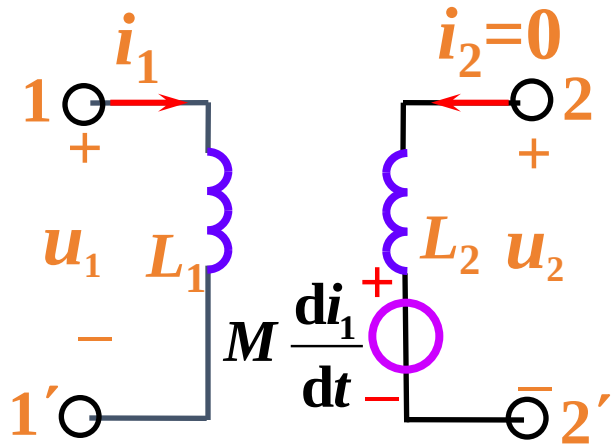
$$u_2 = -M\frac{\mathrm{d}i_1}{\mathrm{d}t}$$



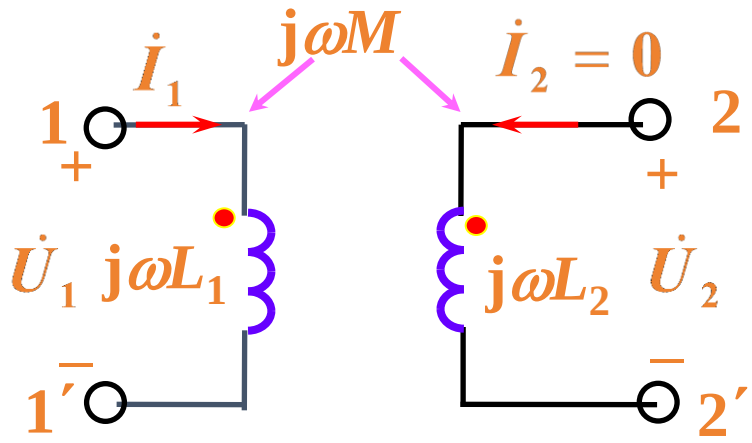
$$u_2 = -M\frac{\mathrm{d}i_1}{\mathrm{d}t}$$

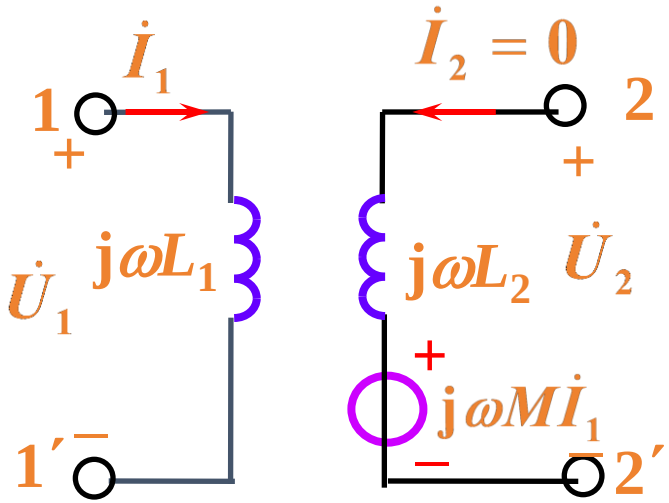
(4) 互感电压用附加的电压源代替





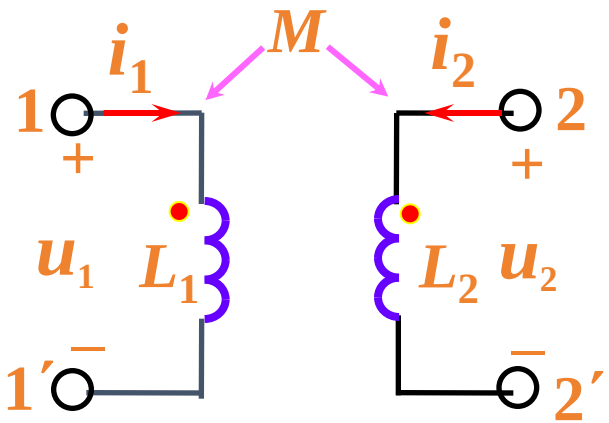
对于正弦稳态电路，则可用相量模型，并依据相量的微分性质

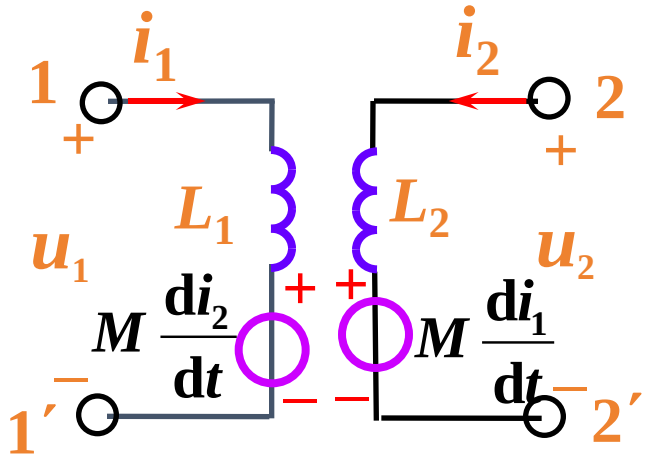




§11-2 耦合电感的VCR 耦合系数

一.耦合电感的VCR

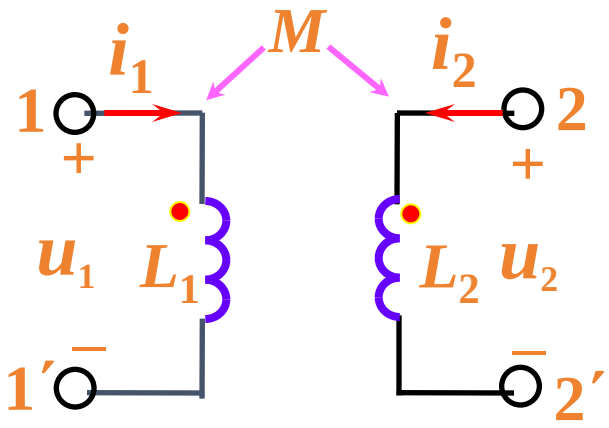


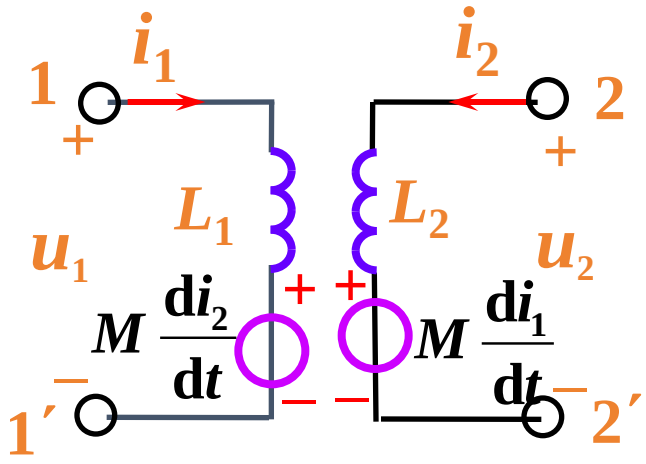


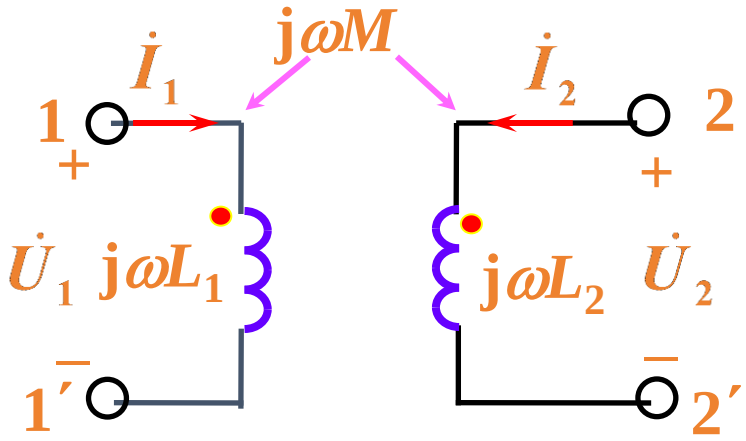
$$u_1 = L_1 \frac{\mathrm{d}i_1}{\mathrm{d}t} + M \frac{\mathrm{d}i_2}{\mathrm{d}t}$$

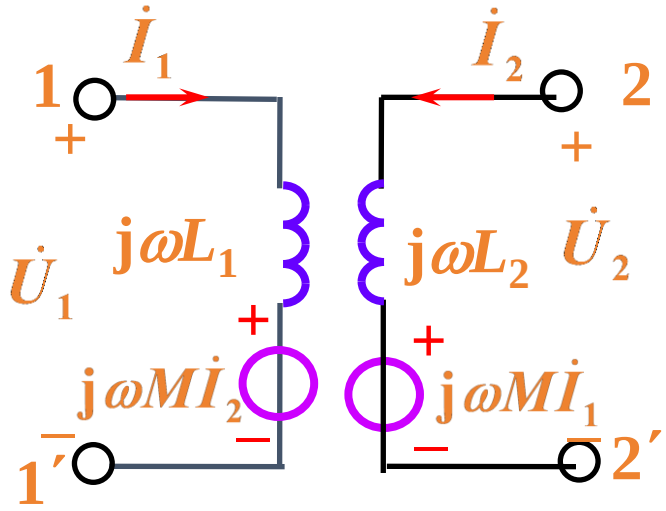
$$u_2 = L_2 \frac{\mathrm{d}i_2}{\mathrm{d}t} + M \frac{\mathrm{d}i_1}{\mathrm{d}t}$$

一.耦合电感的VCR





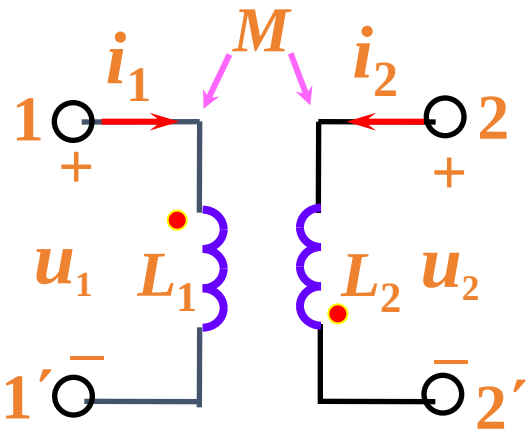


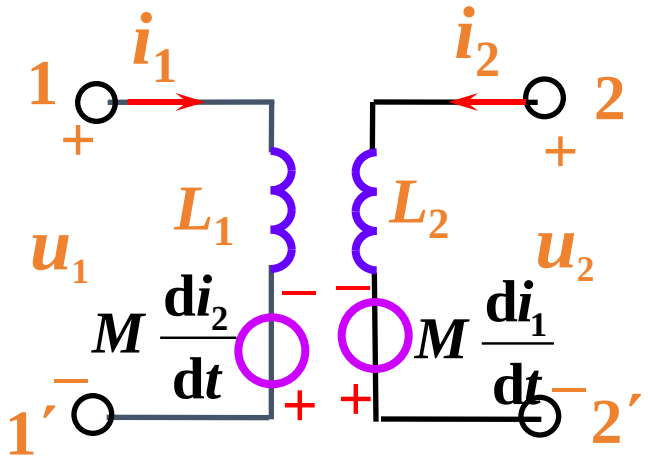


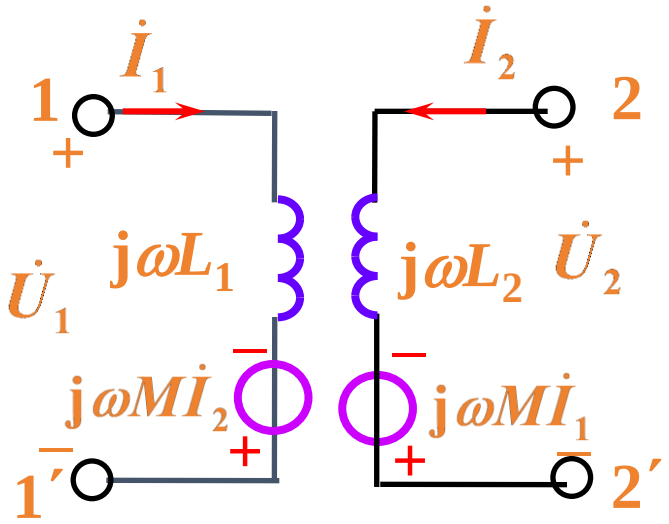
$$\dot{U}_1 = \mathrm{j}\omega L_1 \dot{I}_1 + \mathrm{j}\omega M \dot{I}_2$$

$$\dot{U}_2 = \mathrm{j}\omega L_2 \dot{I}_2 + \mathrm{j}\omega M \dot{I}_1$$

一.耦合电感的VCR







$$u_1 = L_1\frac{\mathrm{d}i_1}{\mathrm{d}t} - M\frac{\mathrm{d}i_2}{\mathrm{d}t}$$

$$u_2 = L_2\frac{\mathrm{d}i_2}{\mathrm{d}t} - M\frac{\mathrm{d}i_1}{\mathrm{d}t}$$

$$\dot{U}_1 = \mathrm{j}\omega L_1\dot{I}_1 - \mathrm{j}\omega M\dot{I}_2$$

$$\dot{U}_2 = \mathrm{j}\omega L_2\dot{I}_2 - \mathrm{j}\omega M\dot{I}_1$$

[例] 求图示电路中的开路电压$\dot{U}_{ab}$。

解：$\dot{U}_{ab} = j5\,\dot{I}_2 + j2\dot{I}_1$

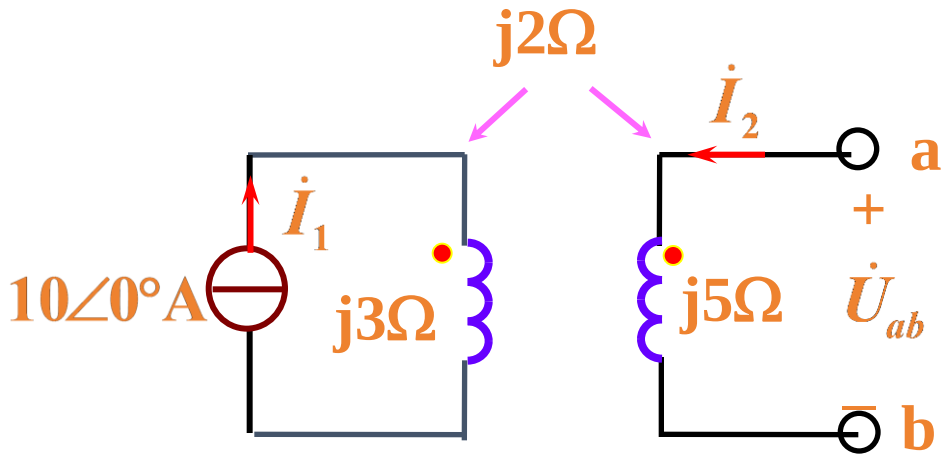
$\dot{I}_2 = 0$

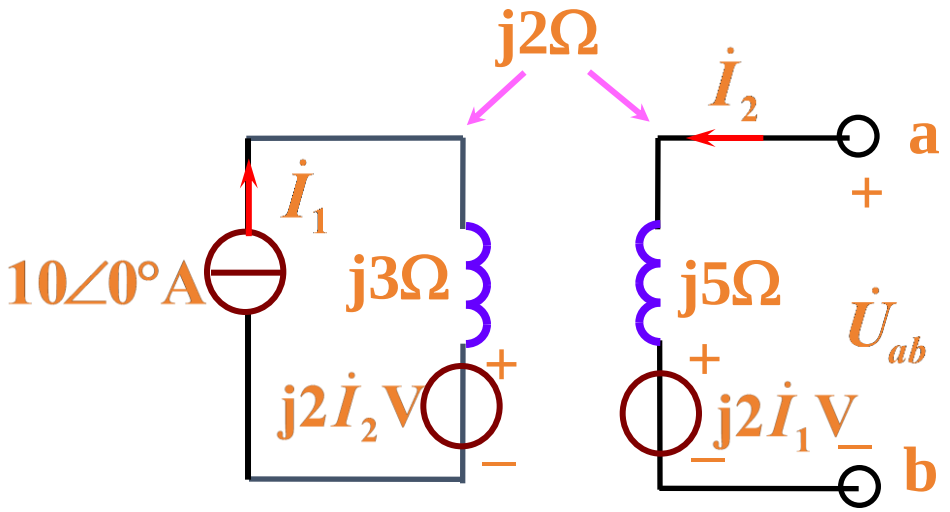
$\dot{U}_{ab} = j2\dot{I}_1$

$= j2 \times 10\angle 0°$

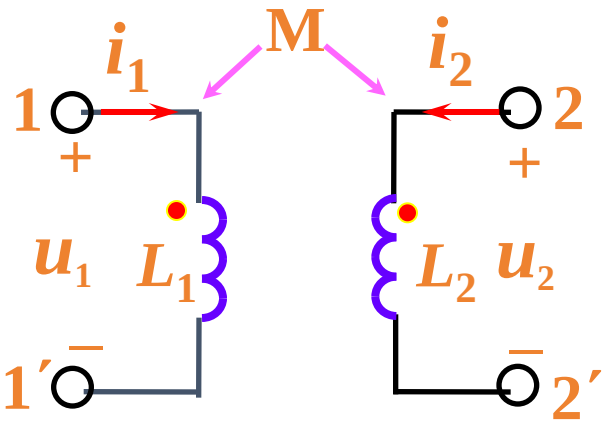
$= j20\text{V}$

$= 20\angle 90°\ \text{V}$





二、 耦合系数



i_1在线圈L_1产生自感磁链 $\Psi_{11}=N_1\Phi_{11}=L_1i_1$
在线圈L_2产生互感磁链 $\Psi_{21}=N_2\Phi_{21}=Mi_1$

i_2 在线圈L_2产生自感磁链 $\Psi_{22}=N_2\Phi_{22}=L_2i_2$
在线圈L_1产生互感磁链 $\Psi_{12}=N_1\Phi_{12}=Mi_2$

在极限情况下， $\Phi_{21}=\Phi_{11}$， $\Phi_{12}=\Phi_{22}$，即每一线圈产生的磁通全部与另一线圈交链，这种耦合称为全耦合。

在全耦合时，互感M值最大， $M=M_{max}$，上面的四个关系式(1)和(3)、(2)和(4)两两相乘相等，可推导出：

$$M_{max}=\sqrt{L_1L_2}$$

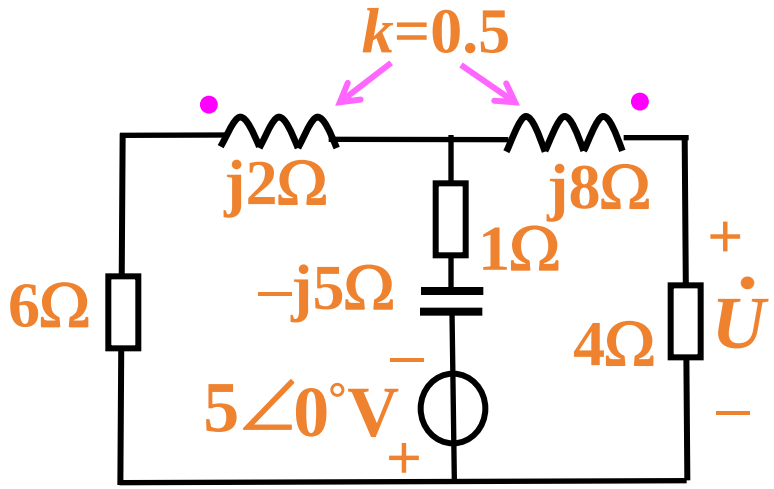
定义：互感M的值与M_{max}的值之比k 称为耦合系数。

$$k=\frac{M}{\sqrt{L_1L_2}}$$ 用来衡量两线圈耦合程度。

$0\le k\le 1$： $k=1$ 全耦合 $k>0.5$ 紧耦合

$k<0.5$ 松耦合 $k=0$ 无耦合

[例] 图示正弦稳态电路中，k为耦合系数，（1）求$\dot{U}$；
（2）求电路消耗的平均功率P。



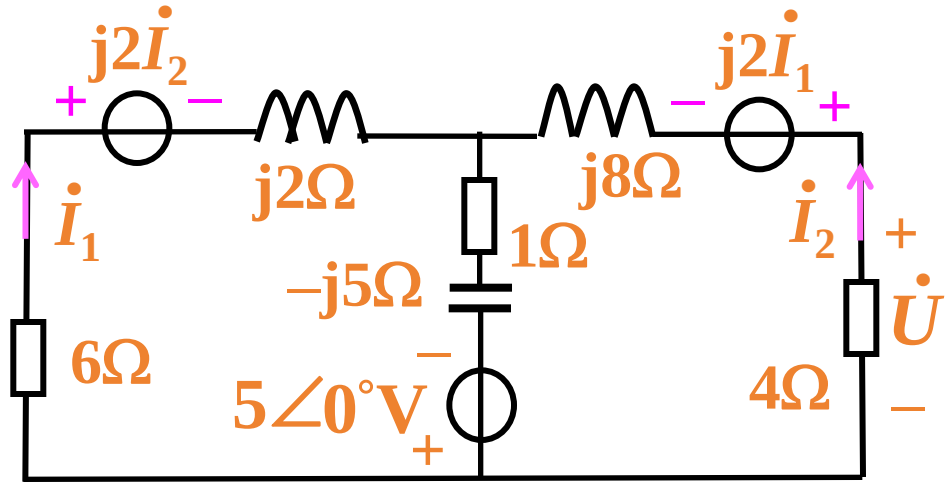
解：$M=k\sqrt{L_1L_2}$

$j\omega M=jk\sqrt{\omega L_1\omega L_2}=j0.5\times4=j2\Omega$

用网孔分析法

$$\begin{cases}(7+j2-j5)\dot{I}_1+(1-j5)\dot{I}_2+j2\dot{I}_2=5\\(1-j5)\dot{I}_1+j2\dot{I}_1+(5+j8-j5)\dot{I}_2=5\end{cases}$$

$$\begin{cases}(7-j3)\dot{I}_1+(1-j3)\dot{I}_2=5\\(1-j3)\dot{I}_1+(5+j3)\dot{I}_2=5\end{cases}$$



$\dot{I}_1=(0.49+j0.46)A$

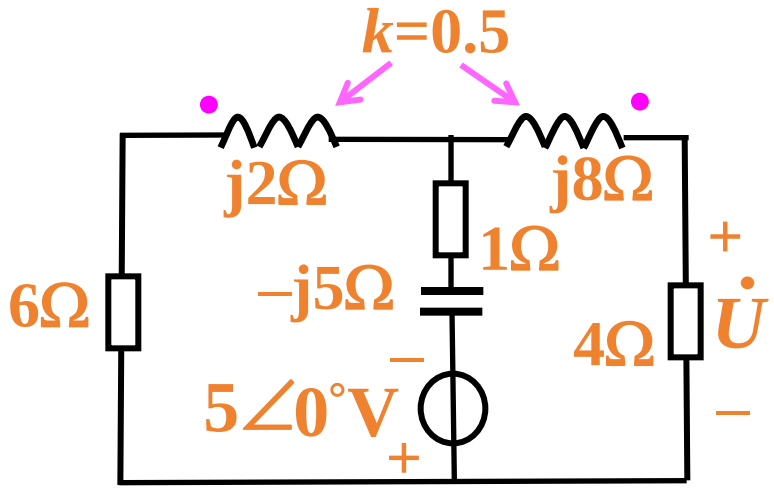
$\dot{I}_2=(0.55-j0.13)A$

[例] 图示正弦稳态电路中，k为耦合系数，（1）求$\dot{U}$；
（2）求电路消耗的平均功率P。

解：$\dot{I}_1=(0.49+j0.46)A$

$\dot{I}_2=(0.55-j0.13)A$

$\dot{U}=-4\dot{I}_2=(-2.2+j0.52)V$



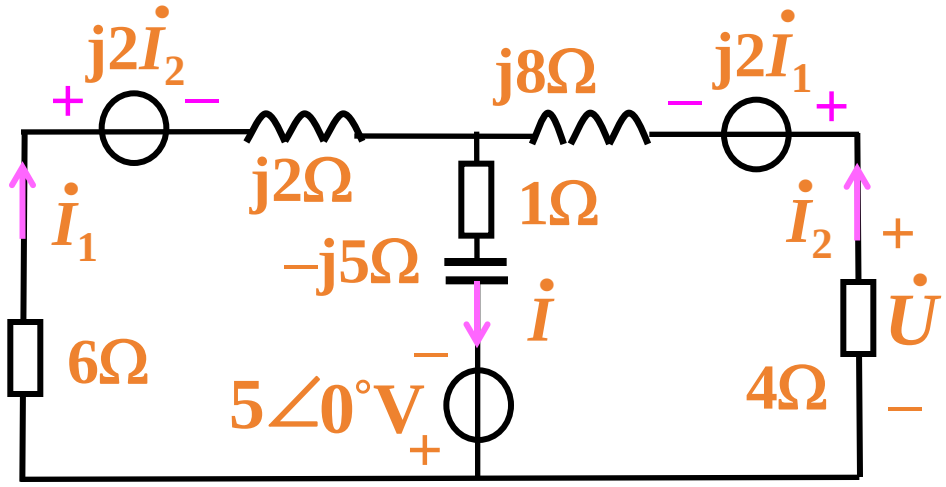
$\dot{I}=\dot{I}_1+\dot{I}_1=(1.04+j0.33)A$

$=1.09\angle 17.6°\ A$

$P=U_S I\cos\varphi$

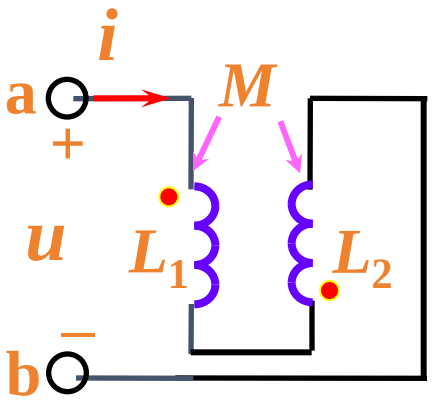
$=5\times 1.09\cos(-17.6°)$

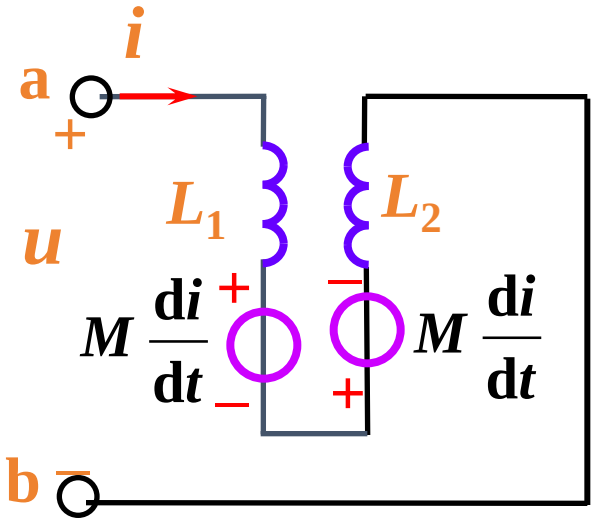
$=5.19W$



三、耦合电感串联联接时的等效电感

1. 顺接串联：异名端相接





$$u_{ab}=L_1\frac{\mathrm{d}i}{\mathrm{d}t}+M\frac{\mathrm{d}i}{\mathrm{d}t}+L_2\frac{\mathrm{d}i}{\mathrm{d}t}+M\frac{\mathrm{d}i}{\mathrm{d}t}=(L_1+L_2+2M)\frac{\mathrm{d}i}{\mathrm{d}t}$$

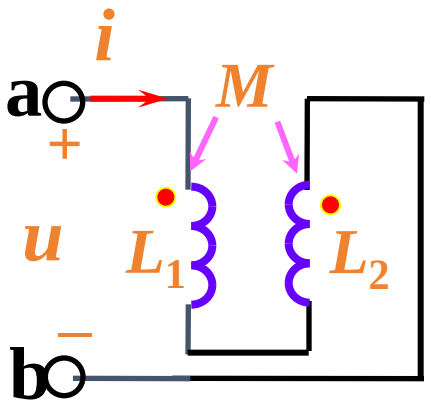
$$u_{ab}=L\frac{\mathrm{d}i}{\mathrm{d}t}$$

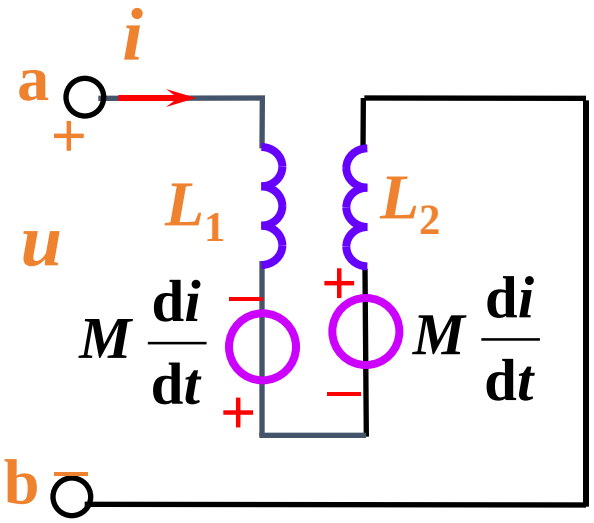
等效电感 $L=L_1+L_2+2M$

正弦稳态时，顺接等效阻抗 $Z=\mathrm{j}\omega(L_1+L_2+2M)$

三、耦合电感串联联接时的等效电感

2. 反接串联：同名端相接





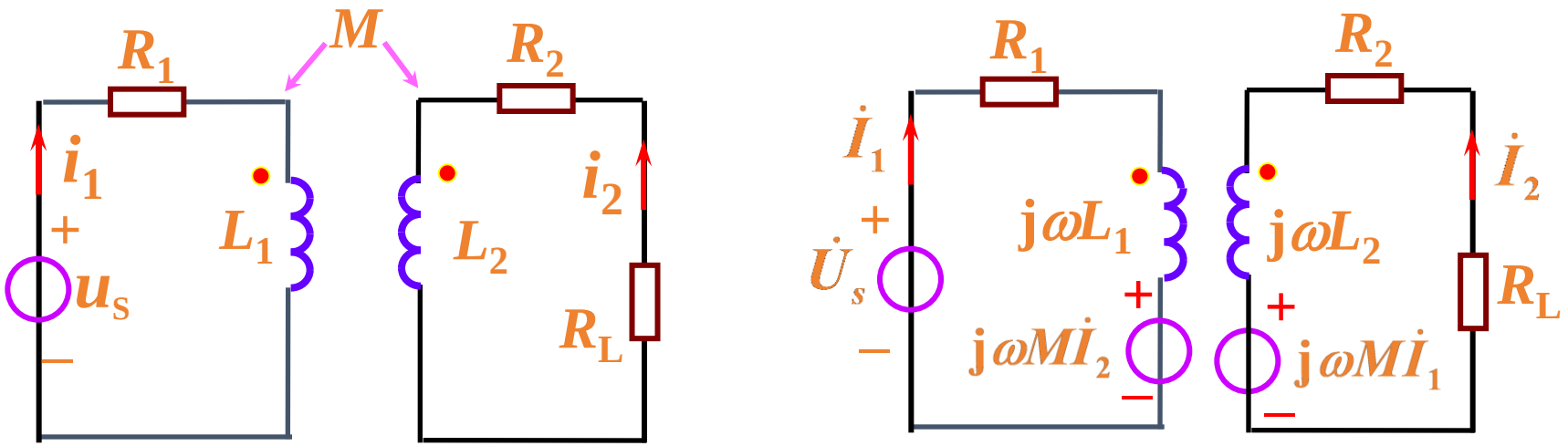
$$u_{ab}=L_1\frac{\mathrm{d}i}{\mathrm{d}t}-M\frac{\mathrm{d}i}{\mathrm{d}t}+L_2\frac{\mathrm{d}i}{\mathrm{d}t}-M\frac{\mathrm{d}i}{\mathrm{d}t}=(L_1+L_2-2M)\frac{\mathrm{d}i}{\mathrm{d}t}$$

$$u_{ab}=L\frac{\mathrm{d}i}{\mathrm{d}t}$$ 等效电感 $L=L_1+L_2-2M$

正弦稳态时，反接等效阻抗 $Z=\mathrm{j}\omega(L_1+L_2-2M)$

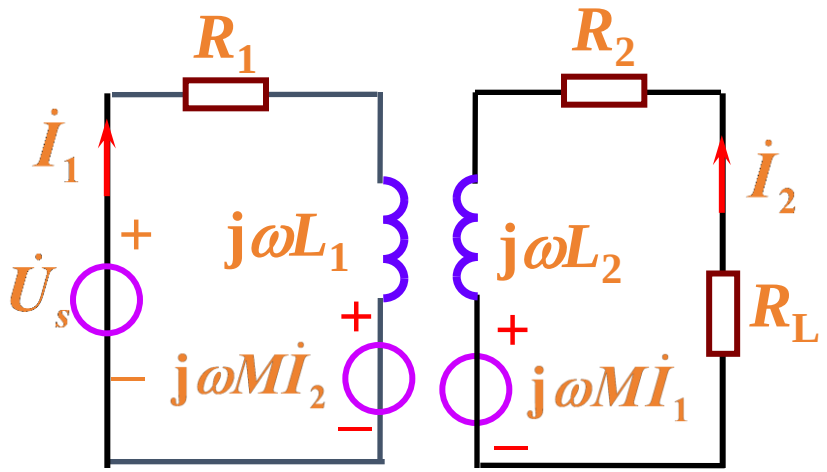
§11-3 空心变压器电路的分析 反映阻抗

空心变压器就是没有铁心的变压器。可以用耦合电感元件来构成它的电路模型，如下图所示。



1.回路法

$$(R_1+\mathrm{j}\omega L_1)\dot{I}_1+\mathrm{j}\omega M\dot{I}_2=\dot{U}_s$$
$$\mathrm{j}\omega M\dot{I}_1+(R_2+R_L+\mathrm{j}\omega L_2)\dot{I}_2=0$$



$$Z_{11}\dot{I}_1+Z_{12}\dot{I}_2=\dot{U}_s$$
$$Z_{21}\dot{I}_1+Z_{22}\dot{I}_2=0$$

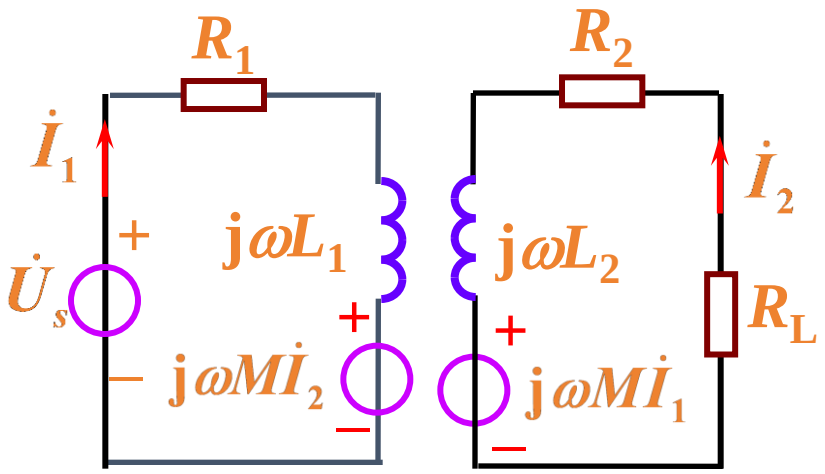
$$Z_{11}=R_1+\mathrm{j}\omega L_1$$
$$Z_{22}=R_2+R_L+\mathrm{j}\omega L_2$$
$$Z_{12}=Z_{21}=\mathrm{j}\omega M$$

依据克莱姆法则

$$\dot{I}_1=\frac{\begin{vmatrix}\dot{U}_s & Z_{12}\\ 0 & Z_{22}\end{vmatrix}}{\begin{vmatrix}Z_{11} & Z_{12}\\ Z_{21} & Z_{22}\end{vmatrix}}=\frac{Z_{22}\dot{U}_s}{Z_{11}Z_{22}-Z_{12}Z_{21}}$$

$$\dot{I}_2=\frac{\begin{vmatrix}Z_{11} & \dot{U}_s\\ Z_{21} & 0\end{vmatrix}}{\begin{vmatrix}Z_{11} & Z_{12}\\ Z_{21} & Z_{22}\end{vmatrix}}=\frac{-Z_{21}\dot{U}_s}{Z_{11}Z_{22}-Z_{12}Z_{21}}$$

$$\dot{I}_1=\frac{Z_{22}\dot{U}_s}{Z_{11}Z_{22}-Z_{12}Z_{21}}$$



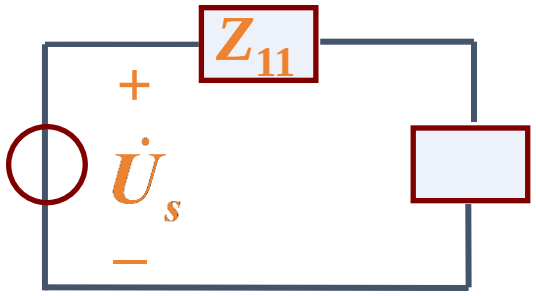
$Z_{11}=R_1+\mathrm{j}\omega L_1$

$Z_{22}=R_2+R_\mathrm{L}+\mathrm{j}\omega L_2$

$Z_{12}=Z_{21}=\mathrm{j}\omega M$

初级看进去的等效阻抗

$$Z_\mathrm{i}=\frac{\dot{U}_s}{\dot{I}_1}=\frac{Z_{11}Z_{22}-Z_{12}Z_{21}}{Z_{22}}=Z_{11}-\frac{Z_{12}Z_{21}}{Z_{22}}=Z_{11}+\frac{\omega^2M^2}{Z_{22}}$$



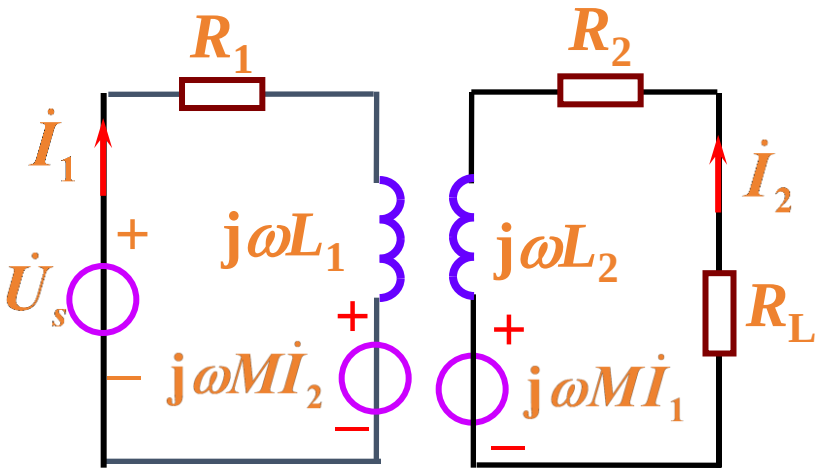
$\dfrac{\omega^2M^2}{Z_{22}}$

初级回路

$\dfrac{\omega^2M^2}{Z_{22}}$ 称为次级回路在初级回路的反映阻抗

2.用反映阻抗计算

$$Z_{11}=R_1+\mathrm{j}\omega L_1$$
$$Z_{22}=R_2+R_L+\mathrm{j}\omega L_2$$
$$Z_{12}=Z_{21}=\mathrm{j}\omega M$$

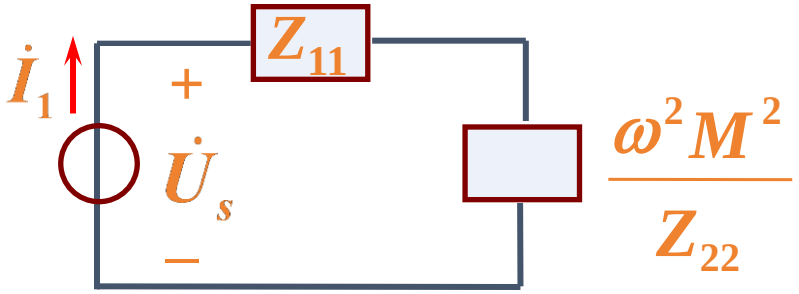


$$\dot{I}_1=\frac{\dot{U}_s}{Z_{11}+\dfrac{\omega^2M^2}{Z_{22}}}$$

次级电流和初级电流的关系

$$\mathrm{j}\omega M\dot{I}_1+(R_2+R_L+\mathrm{j}\omega L_2)\dot{I}_2=0$$

$$\dot{I}_2=\frac{-\mathrm{j}\omega M\dot{I}_1}{Z_{22}}$$



初级回路

3.用戴维南定理分析

将R_L断开，　$\dot{I}_2 = 0$

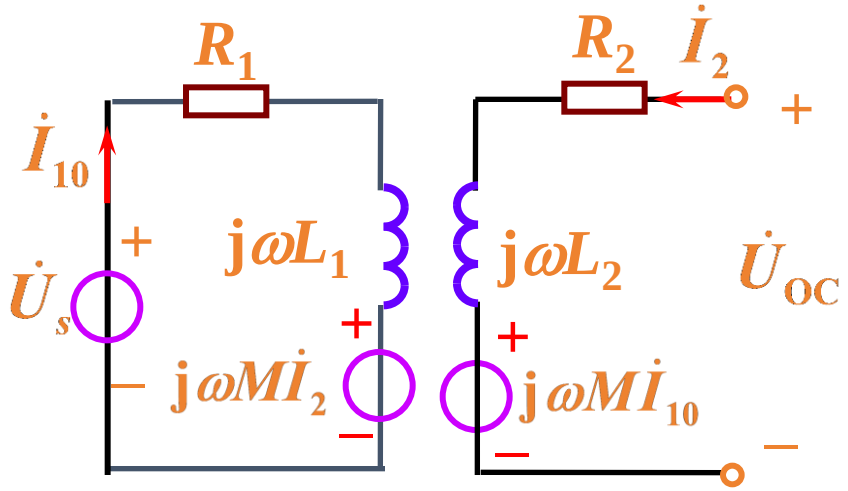
$$\dot{I}_{10} = \frac{\dot{U}_s}{R_1 + \mathrm{j}\omega L_1} \qquad \dot{U}_{OC} = \mathrm{j}\omega M\dot{I}_{10}$$

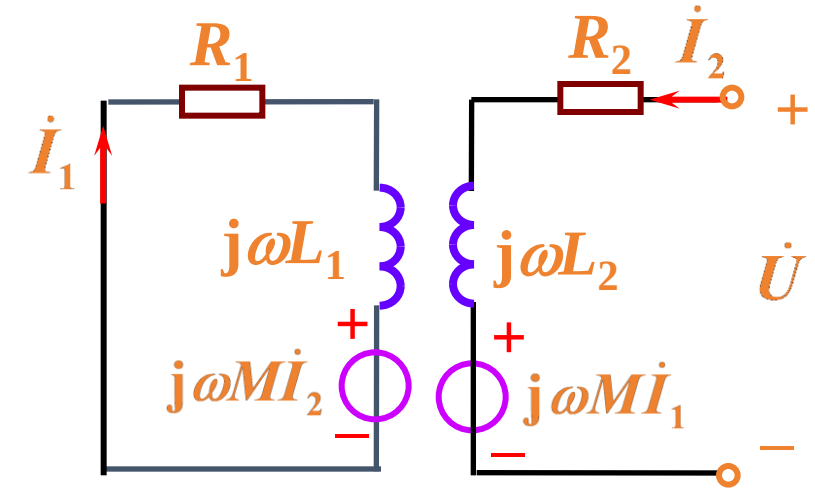
将u_S置零，在开路处外加电压源，可等效看作初级与次级颠倒。

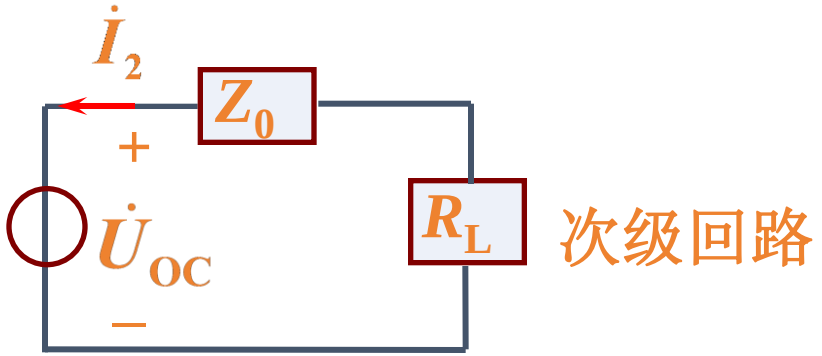
$\dfrac{\omega^2 M^2}{Z_{11}}$ 为初级回路在次级回路的反映阻抗

$$Z_0 = R_2 + \mathrm{j}\omega L_2 + \frac{\omega^2 M^2}{Z_{11}}$$

$$\dot{I}_2 = \frac{-\mathrm{j}\omega M\dot{I}_{10}}{R_2 + \mathrm{j}\omega L_2 + \dfrac{\omega^2 M^2}{Z_{11}} + R_L}$$







次级回路

4. 最大功率传递问题

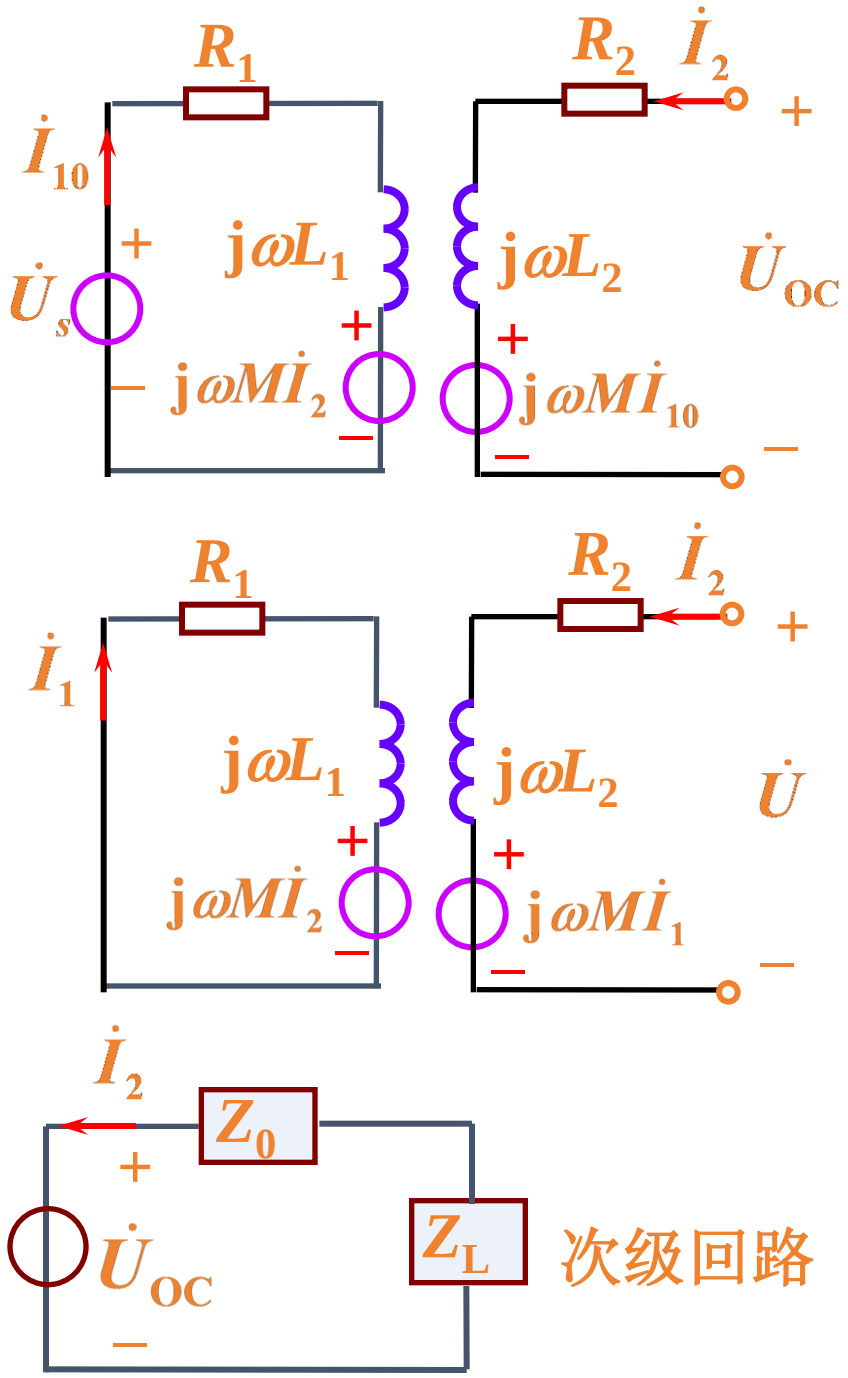
将R_L断开，　$\dot{I}_2 = 0$

$$\dot{I}_{10} = \frac{\dot{U}_s}{R_1 + \mathrm{j}\omega L_1} \qquad \dot{U}_{OC} = \mathrm{j}\omega M\dot{I}_{10}$$

将u_S置零，在开路处外加电压源，可等效看作初级与次级颠倒。

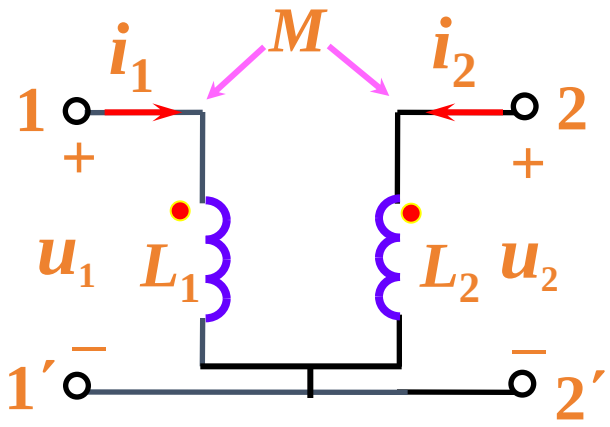
$\dfrac{\omega^2 M^2}{Z_{11}}$ 为初级回路在次级回路的反映阻抗

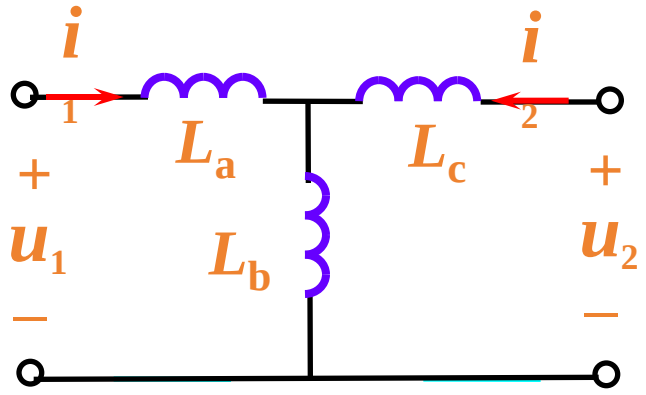
$$Z_0 = R_2 + \mathrm{j}\omega L_2 + \frac{\omega^2 M^2}{Z_{11}}$$



共轭匹配时，即 $Z_L = Z_0^*$ 时，Z_L能获得最大功率。

§11-4 耦合电感的去耦等效电路





$$u_1 = L_1\frac{\mathrm{d}i_1}{\mathrm{d}t} + M\frac{\mathrm{d}i_2}{\mathrm{d}t}$$

$$u_2 = L_2\frac{\mathrm{d}i_2}{\mathrm{d}t} + M\frac{\mathrm{d}i_1}{\mathrm{d}t}$$

$$u_1 = L_a\frac{\mathrm{d}i_1}{\mathrm{d}t} + L_b\frac{\mathrm{d}(i_1+i_2)}{\mathrm{d}t}$$

$$u_2 = L_c\frac{\mathrm{d}i_2}{\mathrm{d}t} + L_b\frac{\mathrm{d}(i_1+i_2)}{\mathrm{d}t}$$

$$u_1 = (L_a + L_b)\frac{\mathrm{d}i_1}{\mathrm{d}t} + L_b\frac{\mathrm{d}i_2}{\mathrm{d}t}$$

$$u_2 = (L_c + L_b)\frac{\mathrm{d}i_2}{\mathrm{d}t} + L_b\frac{\mathrm{d}i_1}{\mathrm{d}t}$$

$L_1 = L_a + L_b$

$L_2 = L_c + L_b$

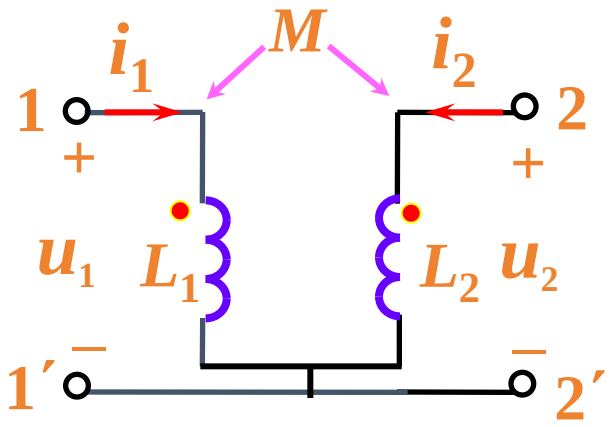
$M = L_b$

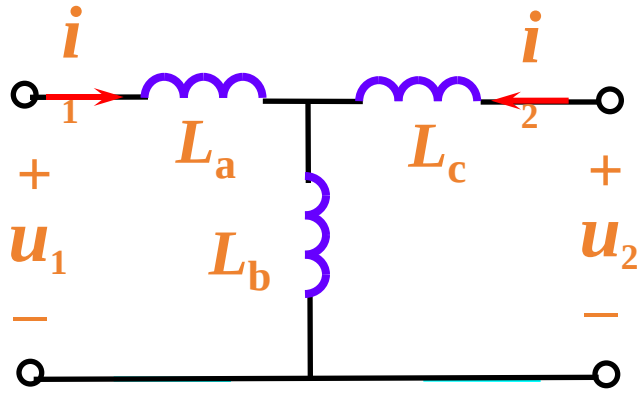
$L_a = L_1 - M$

$L_b = M$

$L_c = L_2 - M$

§11-4 耦合电感的去耦等效电路





$$u_1 = L_1\frac{\mathrm{d}i_1}{\mathrm{d}t} - M\frac{\mathrm{d}i_2}{\mathrm{d}t}$$

$$u_2 = L_2\frac{\mathrm{d}i_2}{\mathrm{d}t} - M\frac{\mathrm{d}i_1}{\mathrm{d}t}$$

$$u_1 = L_a\frac{\mathrm{d}i_1}{\mathrm{d}t} + L_b\frac{\mathrm{d}(i_1+i_2)}{\mathrm{d}t}$$

$$u_2 = L_c\frac{\mathrm{d}i_2}{\mathrm{d}t} + L_b\frac{\mathrm{d}(i_1+i_2)}{\mathrm{d}t}$$

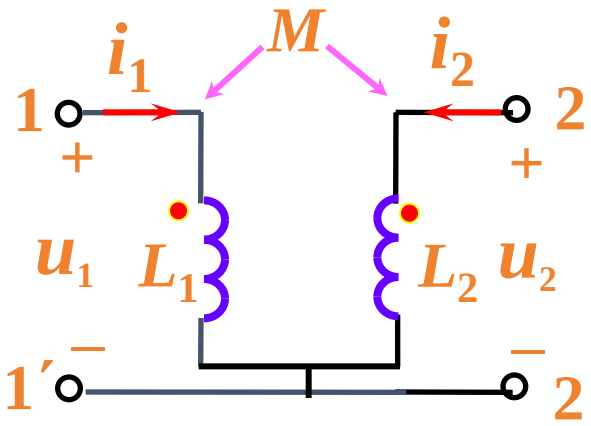
$$u_1 = (L_a+L_b)\frac{\mathrm{d}i_1}{\mathrm{d}t} + L_b\frac{\mathrm{d}i_2}{\mathrm{d}t}$$

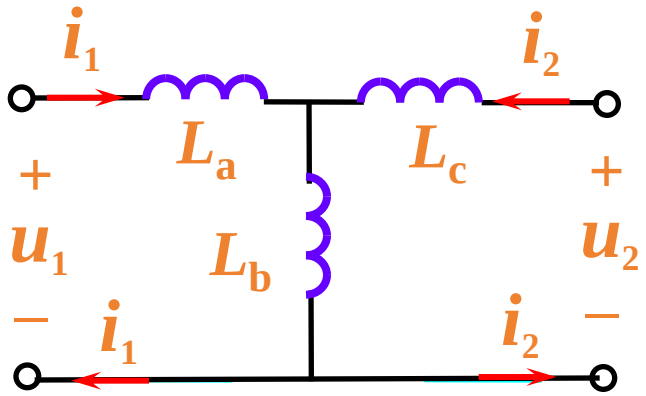
$$u_2 = (L_c+L_b)\frac{\mathrm{d}i_2}{\mathrm{d}t} + L_b\frac{\mathrm{d}i_1}{\mathrm{d}t}$$

$L_1=L_a+L_b$　　$L_a=L_1+M$

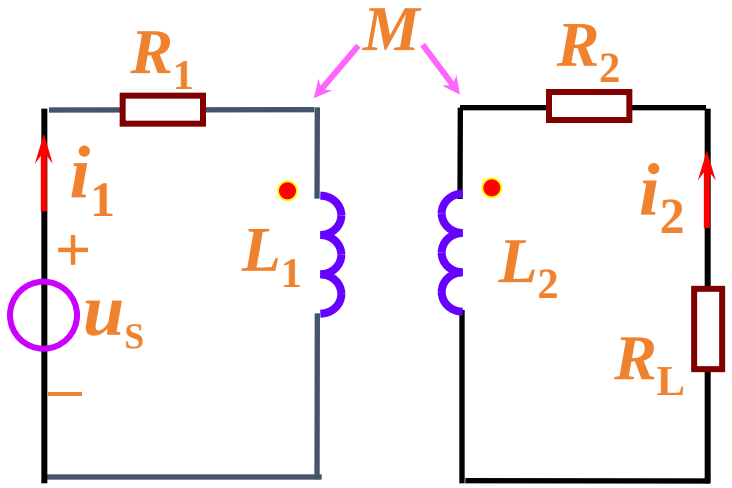
$L_2=L_c+L_b$　　$L_b=-M$

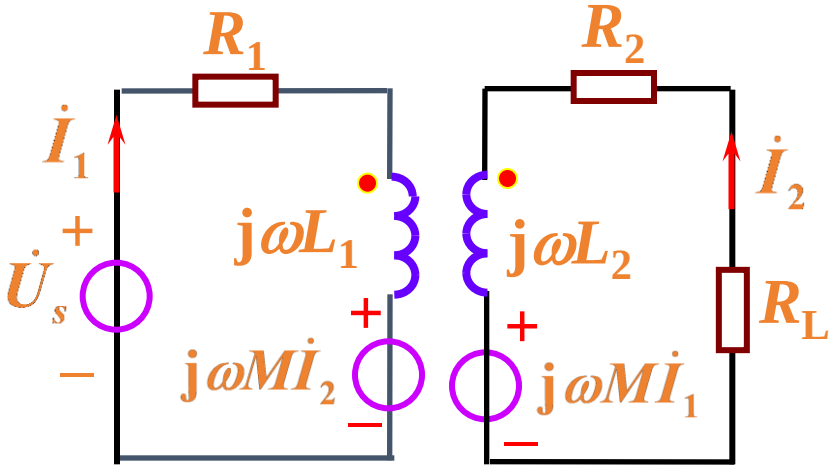
$L_b=-M$　　$L_c=L_2+M$



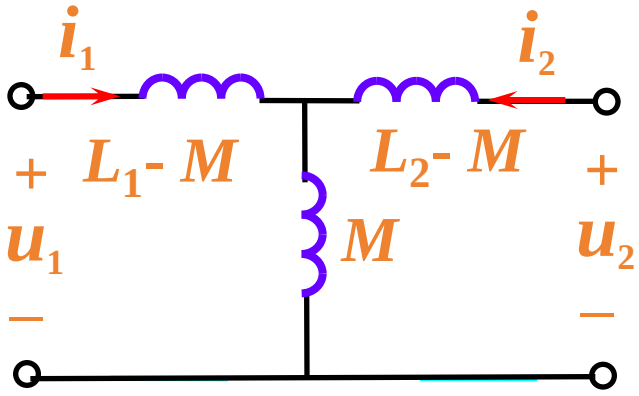


$$L_a = L_1 - M$$
$$L_b = M$$
$$L_c = L_2 - M$$





$$(R_1 + \mathrm{j}\omega L_1)\dot{I}_1 + \mathrm{j}\omega M\dot{I}_2 = \dot{U}_s$$
$$\mathrm{j}\omega M\dot{I}_1 + (R_2 + R_L + \mathrm{j}\omega L_2)\dot{I}_2 = 0$$



§11-5 理想变压器的VCR

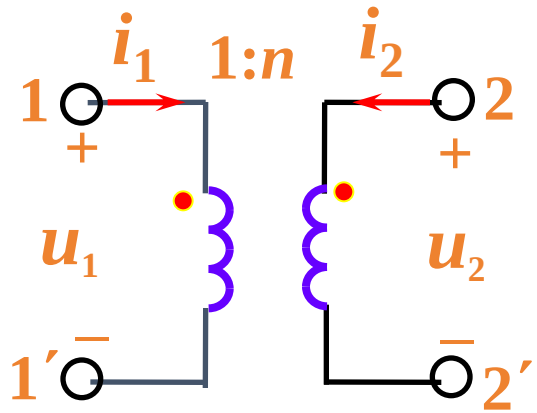
理想变压器是一种双口电阻元件，它也是一种耦合元件，它原是由实际铁心变压器抽象而来的。它的电路模型如图所示。

1.电路模型

理想变压器与耦合电感元件的符号类似，但它唯一的参数只是一个称为变比或匝比的常数n，而没有L_1，L_2和M等参数。

2.理想变压器的VCR

在图中所示同名端和电压、电流的参考方向下



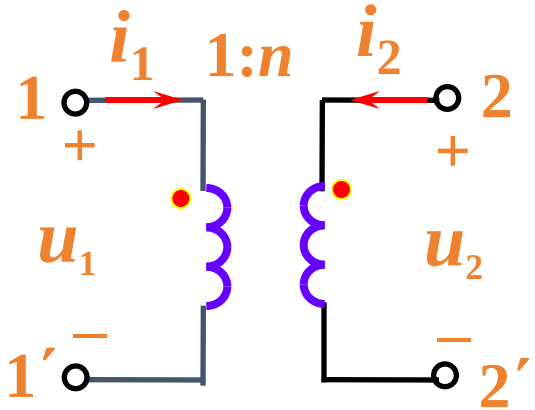
$$u_2 = nu_1$$

$$i_2 = -(1/n)i_1$$

2.理想变压器的VCR

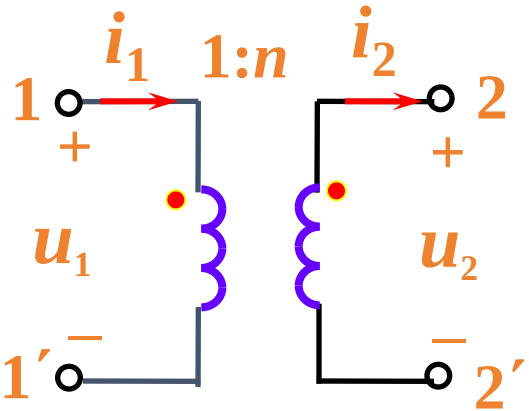
（1）两电压高电位端与同名端一致时，电压比取正，反之取负。

（2）两电流都从同名端流进，电流比取负，反之取正。



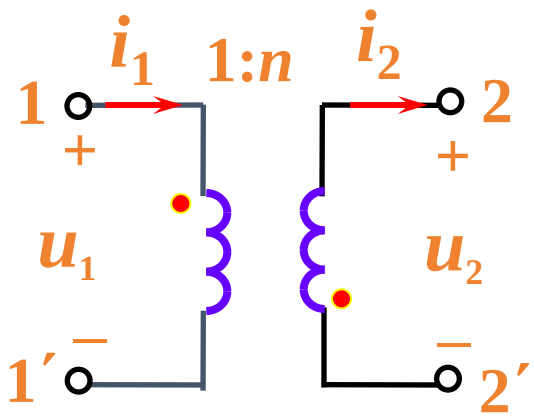
$u_2 = nu_1$

$i_2 = -(1/n)i_1$



$u_2 = nu_1$

$i_1 = n\,i_2$



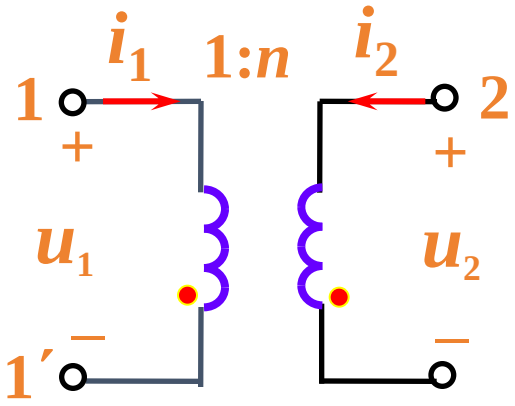
$u_2 = -nu_1$

$i_1 = -ni_2$

2.理想变压器的VCR

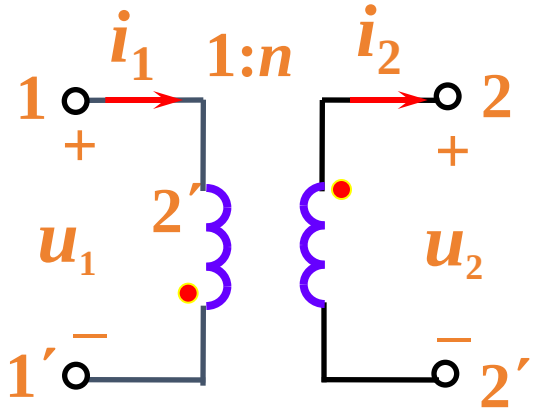
（1）两电压高电位端与同名端一致时，电压比取正，反之取负。

（2）两电流都从同名端流进，电流比取负，反之取正。



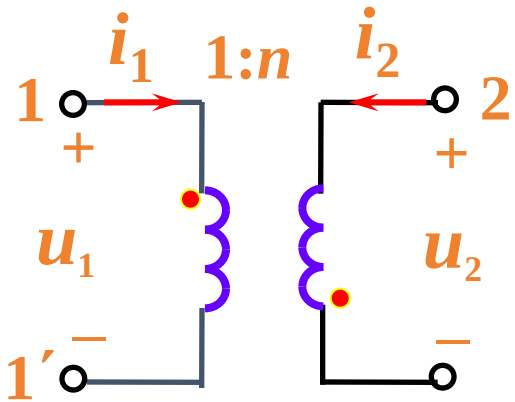
$u_2 = nu_1$

$i_2 = -(1/n)i_1$



$u_2 = -nu_1$

$i_2 = -(1/n)i_1$

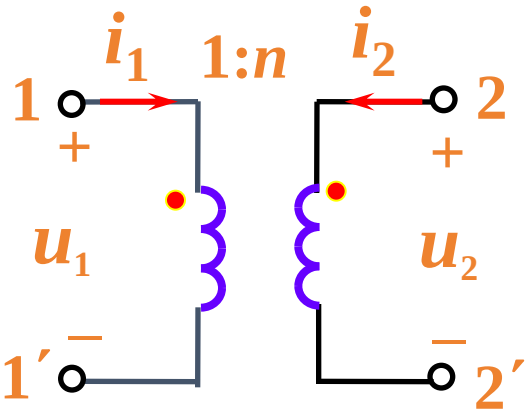


$u_2 = -nu_1$

$i_1 = ni_2$

2.理想变压器的VCR

$u_2=nu_1$　　$i_2=-(1/n)i_1$



在正弦稳态，上面两个VCR式均可表示为相应的相量形式。即

$$\dot{U}_2 = n\dot{U}_1 \qquad \dot{I}_2 = -\frac{1}{n}\dot{I}_1$$

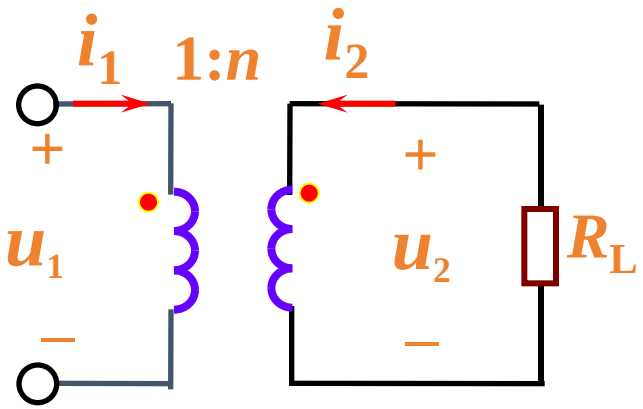
若变压器的初级匝数为N_1，次级匝数为N_2，则匝比（变比）为:

$$n = \frac{N_2}{N_1}$$

3. 功率 $p = u_2 i_2 + u_1 i_1 = nu_1(-\frac{1}{n} i_1) + u_1 i_1 = 0$

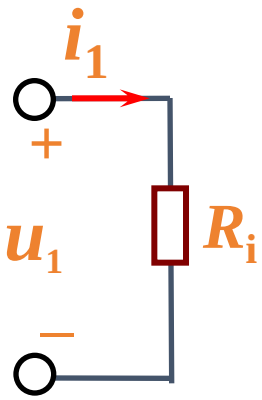
理想变压器既不能消耗能量也不储存能量。

§11-6 理想变压器的阻抗变换性质



$$u_2 = nu_1$$

$$i_2 = -(1/n)i_1$$



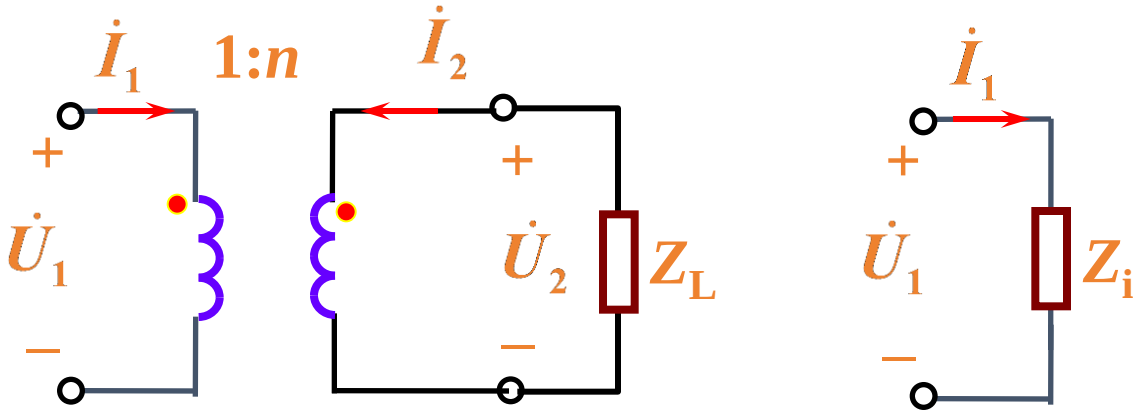
理想变压器不仅可以实现电压变换，电流变换，而且能够实现阻抗变换。

$$R_i = \frac{u_1}{i_1} = \frac{\frac{1}{n}u_2}{-ni_2} = -\frac{1}{n^2}\frac{u_2}{i_2} = \frac{1}{n^2}R_L \qquad \therefore R_i = \frac{1}{n^2}R_L$$

$n>1$ 电阻折合到初级变小 $R_i<R_L$

$n<1$ 电阻折合到初级变大 $R_i>R_L$

在正弦稳态，可由对应的相量模型进行分析



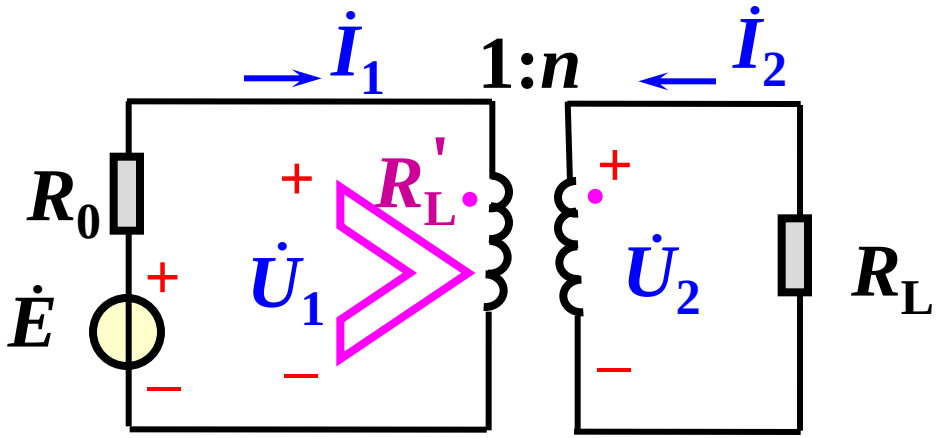
$$Z_i = \frac{\dot{U}_1}{\dot{I}_1} = \frac{\frac{1}{n}\dot{U}_2}{-n\dot{I}_2} = -\frac{1}{n^2}\frac{\dot{U}_2}{\dot{I}_2} = \frac{1}{n^2}Z_L$$

$$Z_i = \frac{1}{n^2}Z_L$$

理想变压器有变换阻抗的性质，可以实现最大功率匹配 。

[例] 电路如图，交流信号源E=120V，内阻R_0=800Ω，负载 R_L=8 Ω，(1) 当R_L折算到原边的等效电阻R'_L= R_0时，求变压器的匝数比和信号源输出的功率；（2）当将负载直接与信号源联接时，信号源输出多大功率？

[解]（1） $R'_L = \frac{1}{n^2}R_L = R_0$



$$\frac{1}{n} = \sqrt{\frac{R_0}{R_L}} = \sqrt{\frac{800}{8}} = 10$$

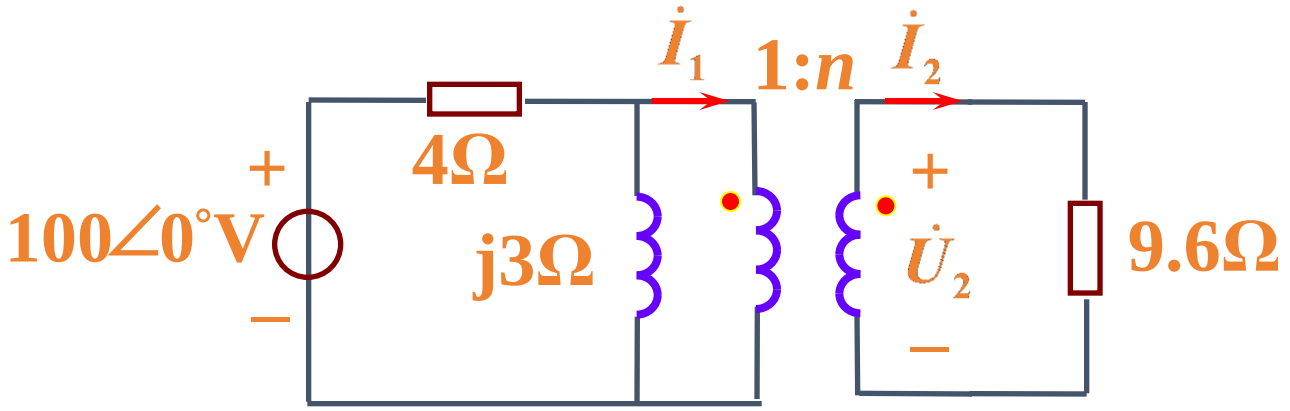
n=0.1

$$P = \left(\frac{E}{R_0 + R'_L}\right)^2 R'_L = \left(\frac{120}{800+800}\right)^2 \times 800 = 4.5\text{W}$$

（2） $P = \left(\frac{120}{800+8}\right)^2 \times 8 = 0.176\text{W}$

可利用变压器进行阻抗匹配

[例]求负载获得最大功率时的匝比n，并求最大功率P_{Lmax} 。



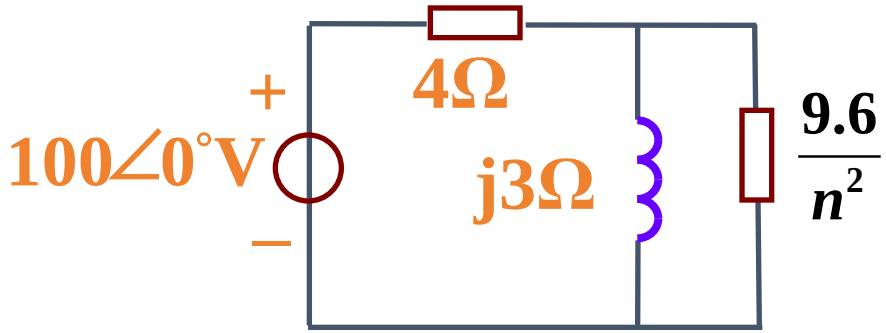
解：将变压器的次级折算到初级

$$Z_0 = \frac{4\times \mathrm{j}3}{4+\mathrm{j}3} = \frac{\mathrm{j}12}{5\angle 36.9^\circ} = 2.4\angle 53.1^\circ\,\Omega$$

由 $Z_i = \dfrac{1}{n^2} Z_L$

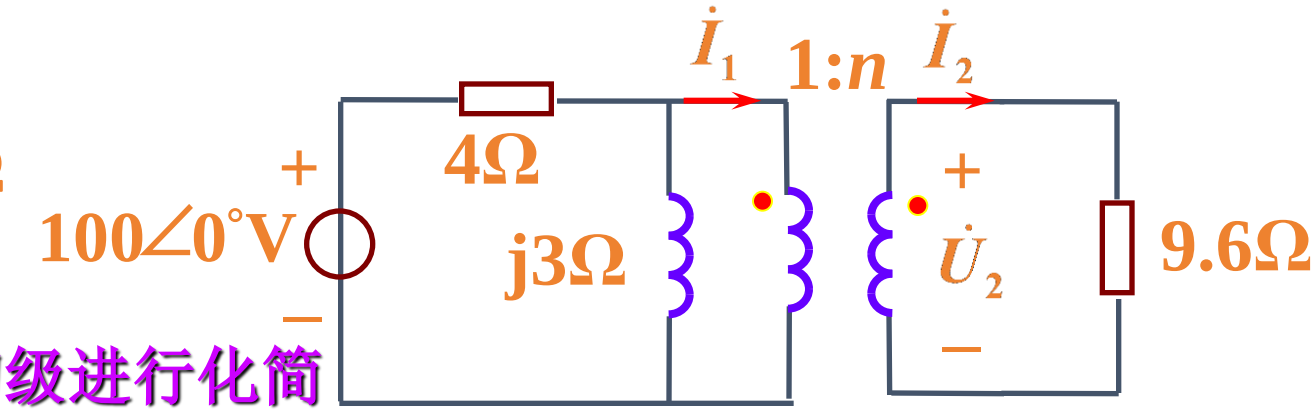
得 $\dfrac{9.6}{n^2} = 2.4$

$n^2 = 4$ $n=2$



$n=2$

$Z_0 = 2.4\angle 53.1^\circ \Omega$

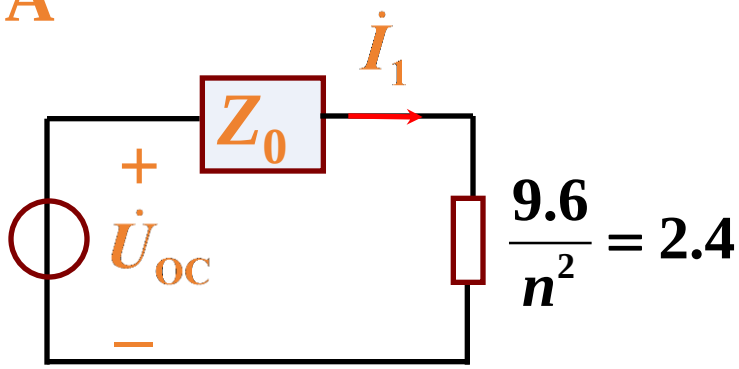


用戴维南定理对初级进行化简

$$\dot{U}_{oc} = \frac{j3}{4+3j} \times 100\angle 0 = 60\angle 53.1^\circ V$$

$$\dot{I}_1 = \frac{60\angle 53.1^\circ}{2.4\angle (53.1)^\circ + 2.4} = 13.986\angle 26.54^\circ \text{ A}$$

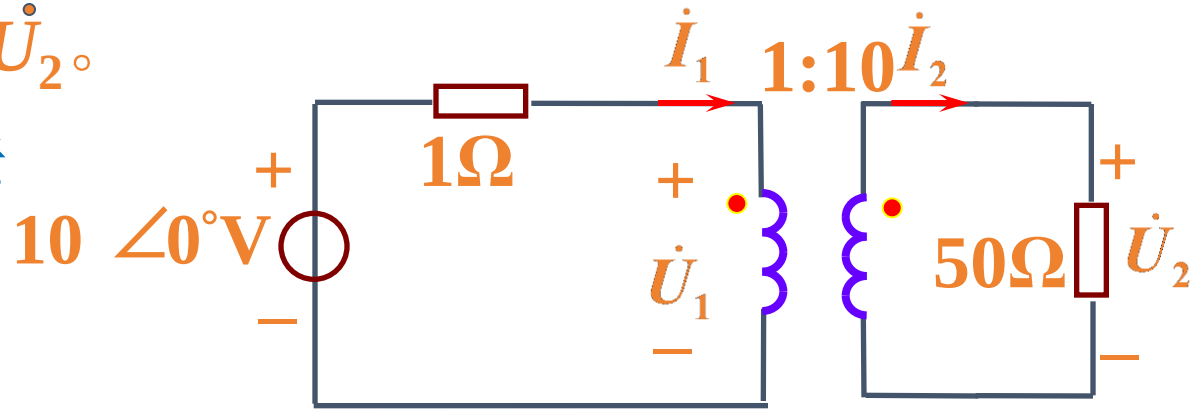
$$P_{L\max} = 13.986^2 \times 2.4 = 496.46 W$$



或 $\dot{I}_2 = \frac{1}{2}\dot{I}_1 = 6.993\angle 26.54^\circ \text{ A}$

$$P_{L\max} = 6.993^2 \times 9.6 = 496.46 W$$

[例] 求图示电路中$\dot{U}_2$。



解：(1)解法1:回路法

$$\dot{U}_2 = 10\dot{U}_1$$

$$\dot{I}_2 = \frac{1}{10}\dot{I}_1$$

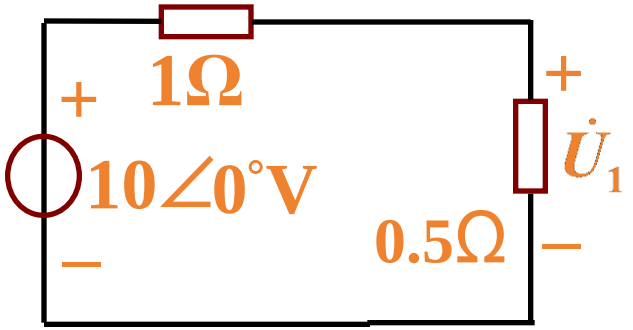
$$\dot{I}_1 + \dot{U}_1 = 10\angle 0^\circ$$

$$50\dot{I}_2 = \dot{U}_2$$

$$\dot{U}_1 = \frac{10}{3}\text{V} \qquad \dot{U}_2 = 10\dot{U}_1 = \frac{100}{3}\text{V}$$

(2)解法2: 把负载阻抗折合到初级 $R_L' = 50/10^2 = 0.5\ \Omega$

$$\dot{U}_1 = \frac{\frac{1}{2}}{1+\frac{1}{2}} \times 10\angle 0^\circ = \frac{10}{3}\text{V}$$



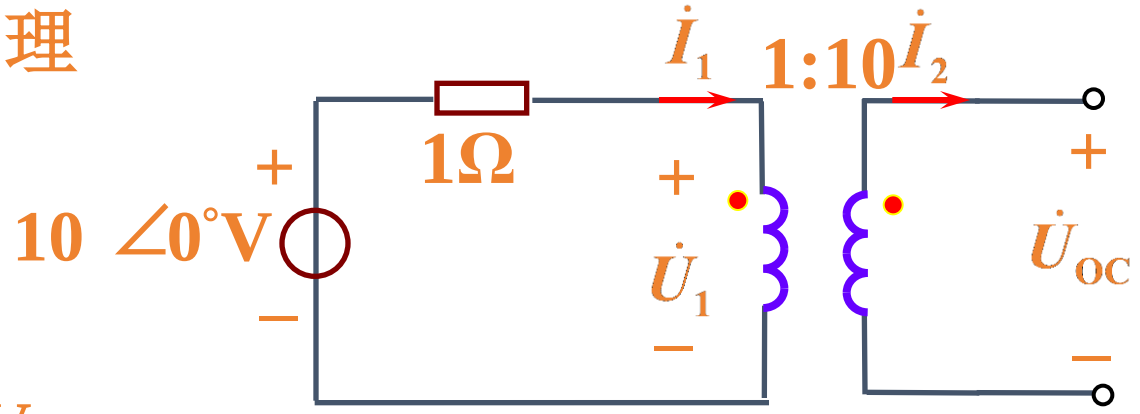
$$\dot{U}_2 = 10\dot{U}_1 = \frac{100}{3}\text{V}$$

(3)解法3: 用戴维南定理

$\dot{I}_2=0$，　$\dot{I}_1=10\dot{I}_2=0$

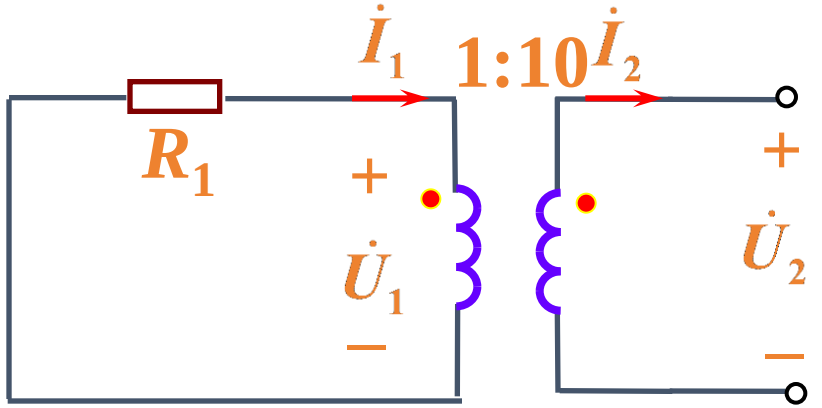
$\dot{U}_1=10\angle 0^\circ \mathrm{V}$

$\dot{U}_{oc}=10\dot{U}_1=100\angle 0^\circ \mathrm{V}$



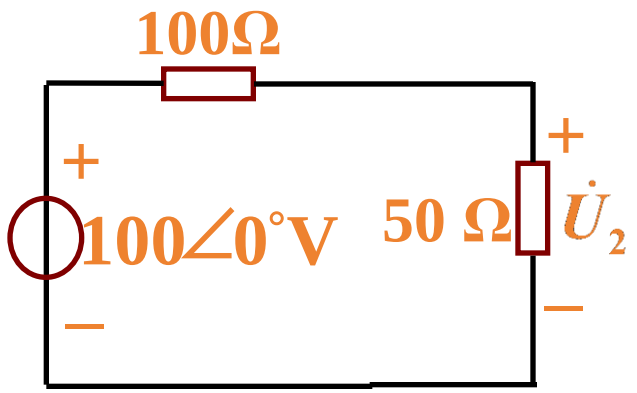
把初级回路阻抗折合到次级回路

$$\frac{\dot{U}_1}{\dot{I}_1}=-R_1$$



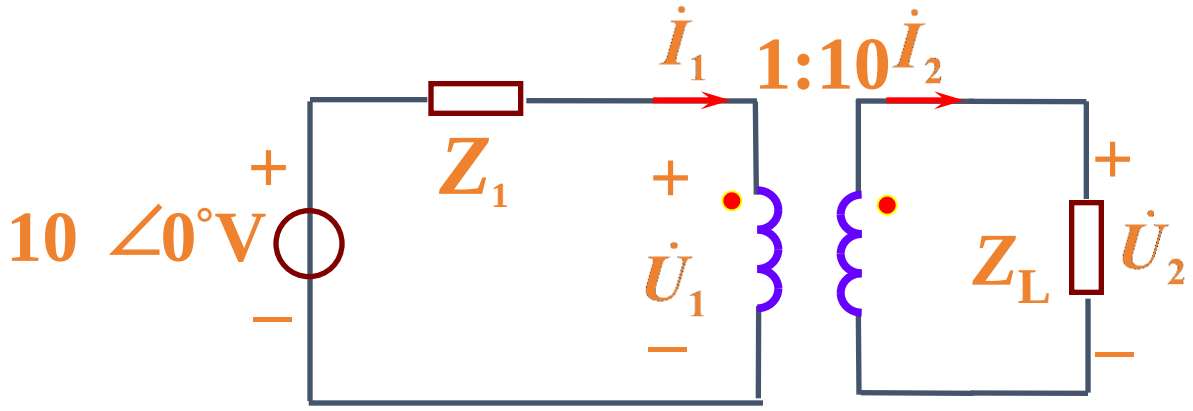
$$R_0=-\frac{\dot{U}_2}{\dot{I}_2}=-\frac{n\dot{U}_1}{\frac{1}{n}\dot{I}_1}=n^2R_1$$

$R_0=10^2\times 1=100\,\Omega$



$$\dot{U}_2=\frac{50}{100+50}\times 100\angle 0^\circ=\frac{100}{3}\mathrm{V}$$

[例]已知Z_1=1+j4Ω，求Z_L为何值时可获最大功率，并求此最大功率。

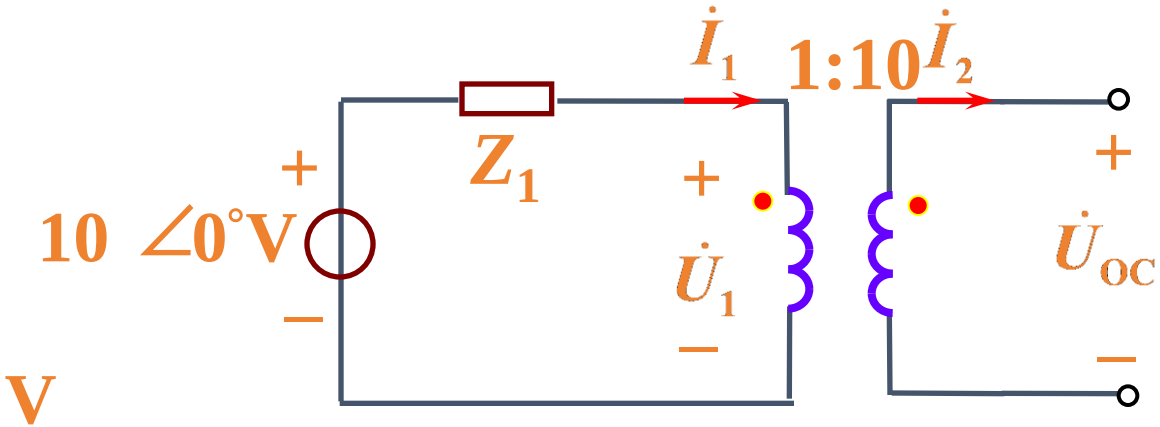


解: 用戴维南定理

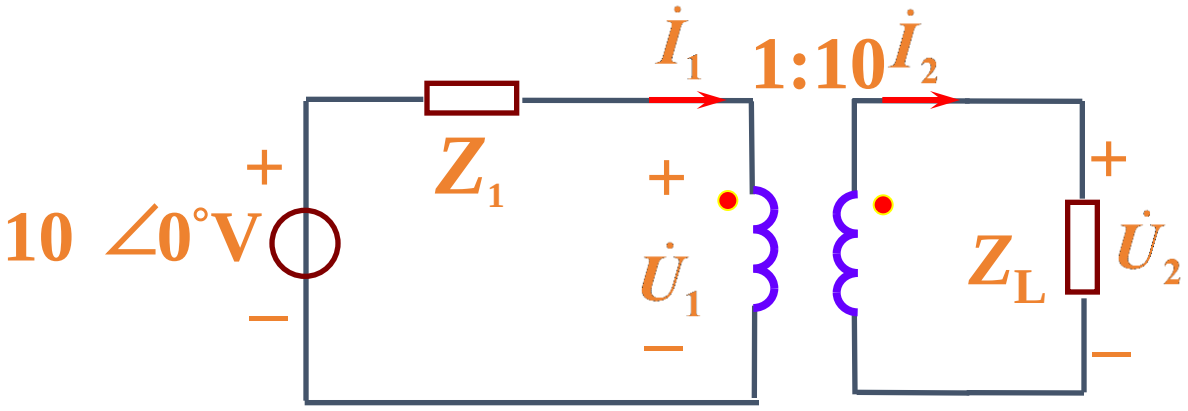
$\dot{I}_2=0$，　$\dot{I}_1=10\dot{I}_2=0$

$\dot{U}_1=10\angle 0^\circ \mathrm{V}$

$\dot{U}_{oc}=10\dot{U}_1=100\angle 0^\circ \mathrm{V}$



[例]已知Z_1=1+j4Ω，求Z_L为何值时可获最大功率，并求此最大功率。

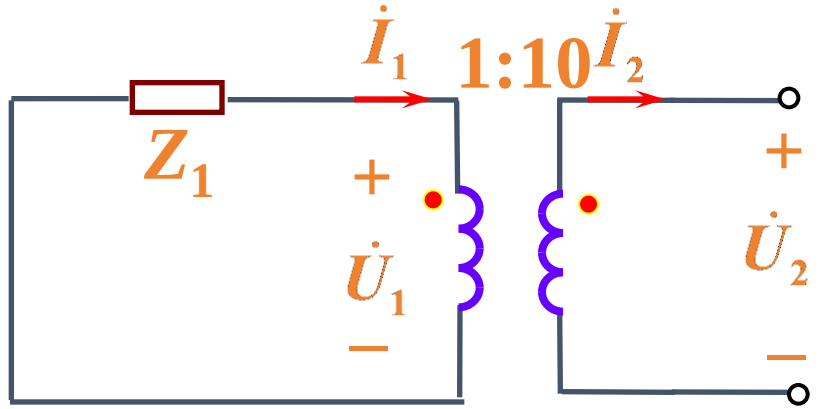


解: 用戴维南定理

把初级回路阻抗折合到次级回路

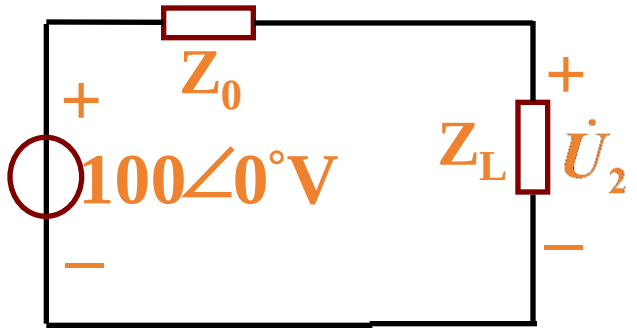
$$\frac{\dot{U}_1}{\dot{I}_1} = -Z_1$$

$$Z_0 = -\frac{\dot{U}_2}{\dot{I}_2} = -\frac{n\dot{U}_1}{\frac{1}{n}\dot{I}_1} = n^2 Z_1$$



Z_0=$10^2 Z_1$=100+j400Ω

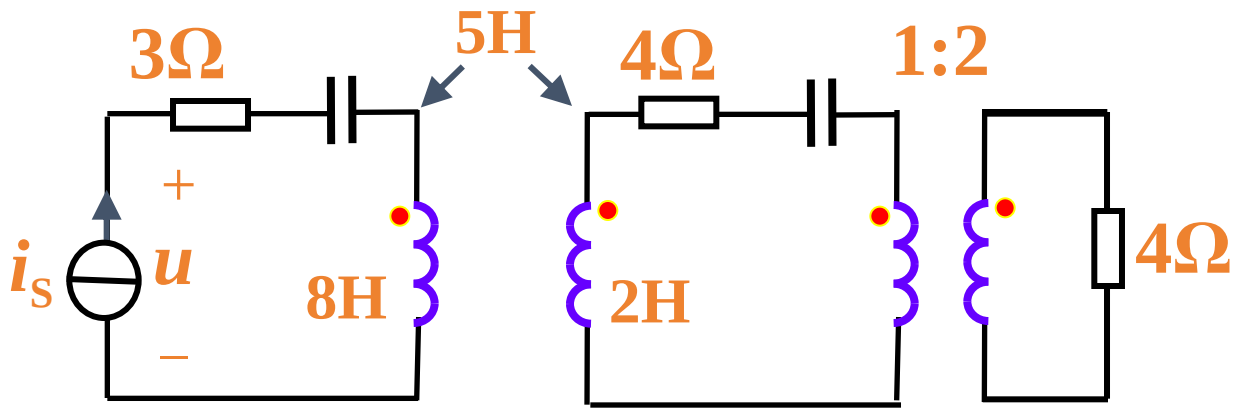
Z_L=Z_0^*=100 −j400Ω时可获最大功率



I_2=100/200=0.5A

P_{max}=100×0.5^2=25W

[例]求图示电路中的电压u(t), 已知i_S(t)=2$\sqrt{2}$costA。



解:

$\dot{I}_S$=2A

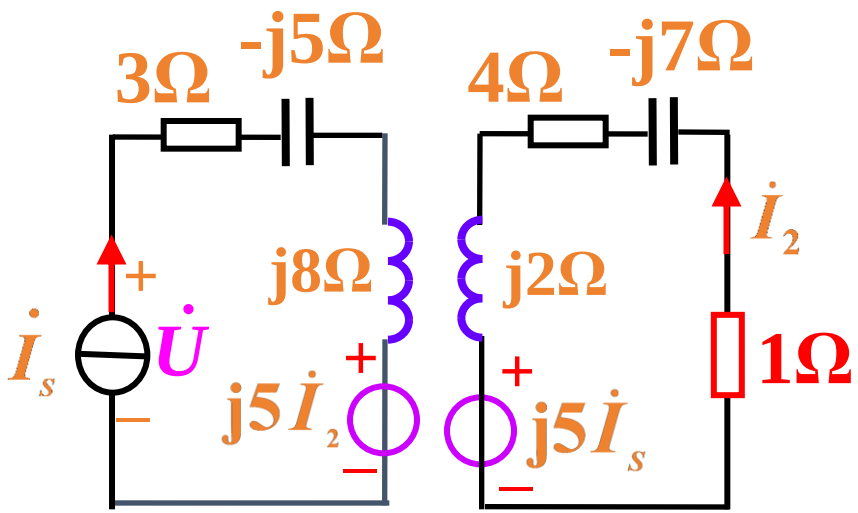
(5+j2-j7)$\dot{I}_2$+j5$\dot{I}_S$=0

$\dot{I}_2$=(1-j)A

$\dot{U}$=(3-j5+j8)$\dot{I}_S$+j5$\dot{I}_2$

=11+j11=11$\sqrt{2}$∠45° V

u(t)=22cos(t+45°)V



相量模型

§11-7 理想变压器的实现

理想变压器是实际变压器的理想化模型，是对互感元件的理想科学抽象，是极限情况下的耦合电感。

理想变压器的三个理想化条件

①无损耗 → 线圈导线无电阻，做芯子的铁磁材料的磁导率无限大。

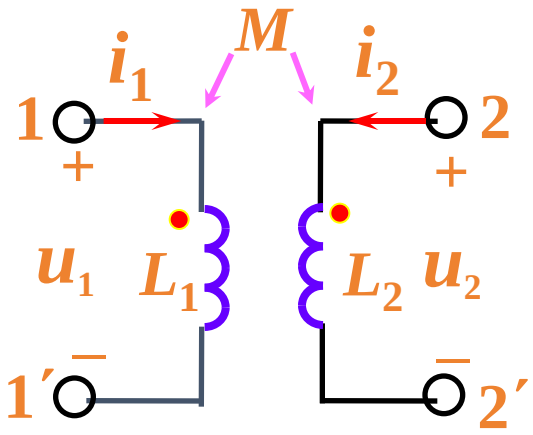
②全耦合 → $k=1 \Rightarrow M=\sqrt{L_1 L_2}$

③参数无限大 → $L_1, L_2, M \Rightarrow \infty$，但 $\sqrt{\dfrac{L_1}{L_2}}=\dfrac{N_1}{N_2}=n$

以上三个条件在工程实际中不可能满足，但在一些实际工程概算中，在误差允许的范围内，把实际变压器当作理想变压器对待，可使计算过程简化。

证明：

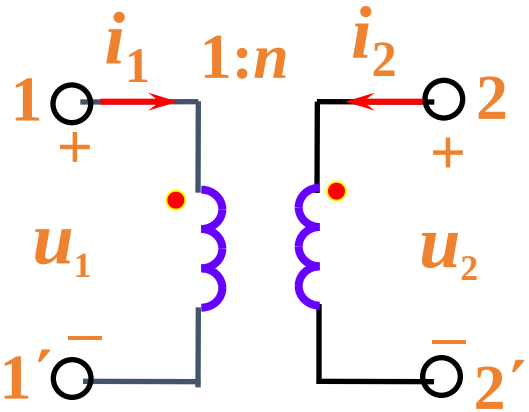
$$u_1 = L_1\frac{\mathrm{d}i_1}{\mathrm{d}t} + M\frac{\mathrm{d}i_2}{\mathrm{d}t} \qquad \frac{u_1}{L_1} = \frac{\mathrm{d}i_1}{\mathrm{d}t} + \frac{M}{L_1}\frac{\mathrm{d}i_2}{\mathrm{d}t}$$



根据：

$$\frac{N_1}{N_2} = \frac{L_1}{M} = \sqrt{\frac{L_1}{L_2}} = \frac{1}{n} \qquad M = \sqrt{L_1 L_2} \qquad n = \frac{N_2}{N_1} \qquad L \to \infty$$

$$\therefore 0 = \frac{\mathrm{d}i_1}{\mathrm{d}t} + \frac{M}{L_1}\frac{\mathrm{d}i_2}{\mathrm{d}t} \qquad \frac{\mathrm{d}i_1}{\mathrm{d}t} = -n\frac{\mathrm{d}i_2}{\mathrm{d}t}$$



两边积分：　$i_1 = -ni_2 + A$

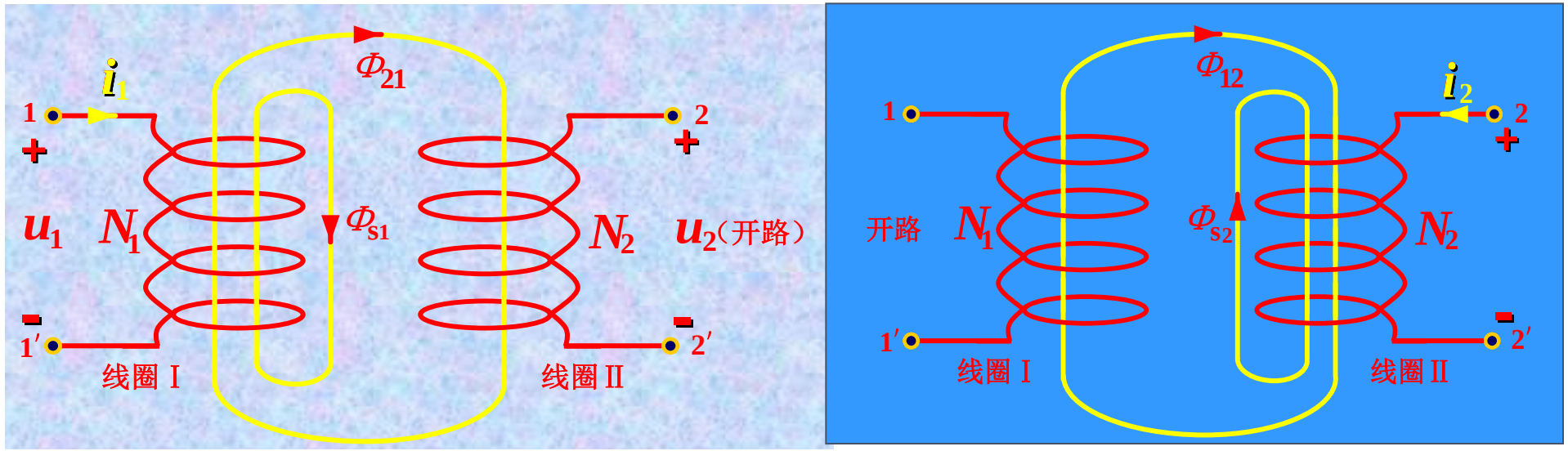
对于时变电压电流：　$A = 0$　　$i_1 = -ni_2$

$$i_2 = -\frac{1}{n}i_1 \qquad u_2 = nu_1$$

耦合电感与理想变压器的关系

(1) 由耦合电感→理想变压器是由储能元件→非能元件，即由动态元件→电阻元件(质的变化)。量变→质变，当$K\rightarrow 1$和L_1、$L_2\rightarrow$无限大时，即可完成这一质变。

(a)为满足 $u_2 = n u_1$ ，必须要求 $K \to 1$，即全耦合。



(a) (b)

当 $\Phi_{S1} = 0$、$\Phi_{S2} = 0$ 时，$\Phi_{11} = \Phi_{21}$、$\Phi_{22} = \Phi_{12}$

穿过N_1、N_2的磁通 Φ ，$\Phi = \Phi_{11} + \Phi_{22}$

$$u_1 = N_1 \frac{\mathrm{d}\Phi}{\mathrm{d}t} \qquad u_2 = N_2 \frac{\mathrm{d}\Phi}{\mathrm{d}t}$$

故得 $u_2 = n u_1$

(b)理想变压器 u_1并不取决于 i_1或 i_2，只取决于 u_2。

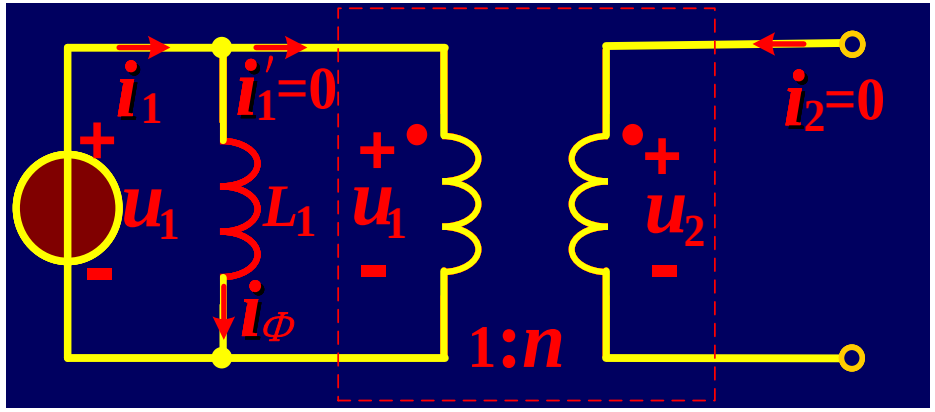
同样，i_1只取决于 i_2，与u_1 、u_2无关。

设想在一次侧施加电压u_1，并使二次侧开路，则$i_2=0$从而i_1也必须为零。在非零u_1下，L_1应$\rightarrow\infty$。类似地可说明$L_2\rightarrow\infty$。

(c)采取技术措施可力争实现上述两条件。

设计精良的变压器可认为是理想的； 一般变压器也可用理想变压器为核心构成它的模型。

(2)全耦合变压器　　只满足条件(a)

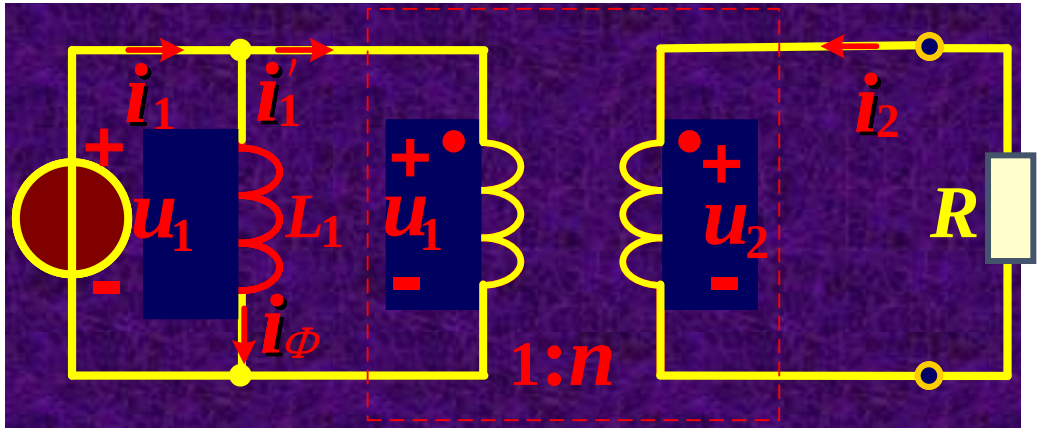


$$K \to 1 \qquad u_2 = nu_1$$

$$i_1 = i_\Phi$$

磁化电流

虚线框——理想变压器

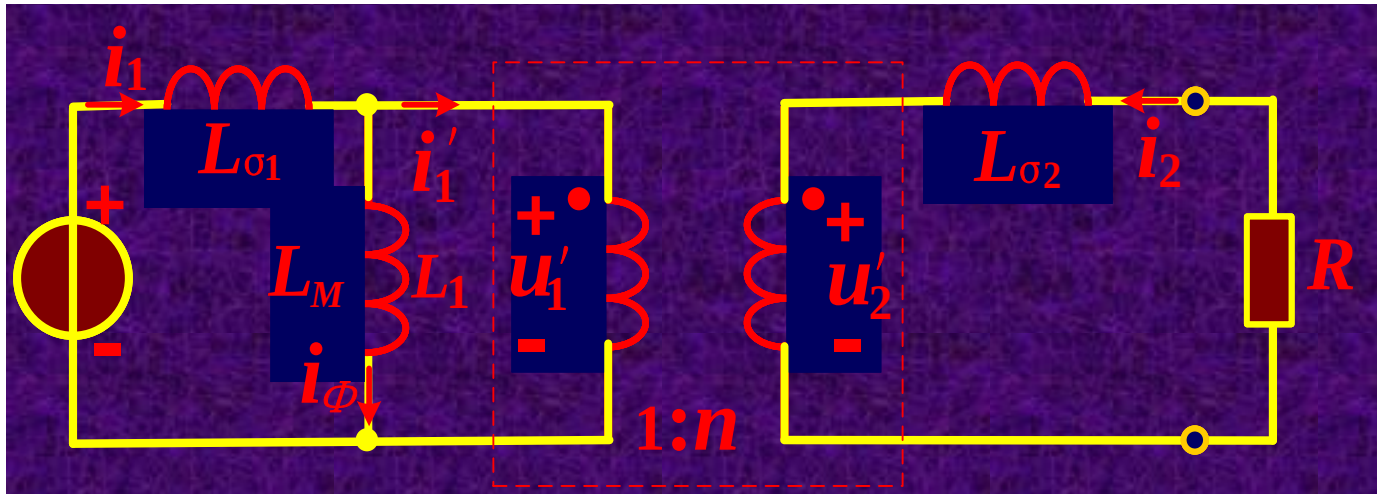


$$u_2 = n\,u_1$$

$$i_1 = i_\Phi + i_1'$$

$$= i_\Phi - ni_2$$

(3)一般变压器　　(a)、(b)两条件均不满足



$L_{\sigma 1}$、$L_{\sigma 2}$　　漏磁通电感

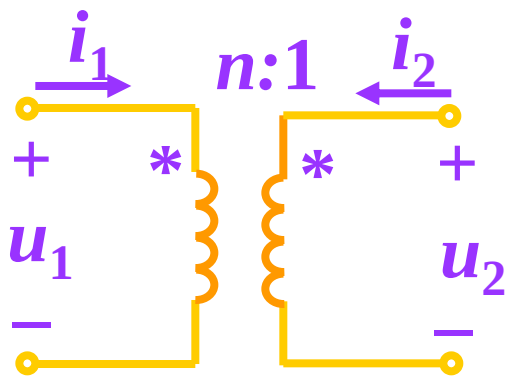
$$L_M = L_1 - L_{\sigma 1}$$　　磁化电感

模型中未计入铜损耗、铁损耗

§11-8 铁心变压器的模型

铁心变压器是常用的器件，实际变压器是有损耗的，也不可能全耦合，$k \neq 1$，且L_1，M，$L_2 \neq \infty$。除了用具有互感的电路来分析计算以外，还常用含有理想变压器的电路模型来表示。

1.理想变压器(全耦合$k=1$，无损，L_1、L_2为无限大)

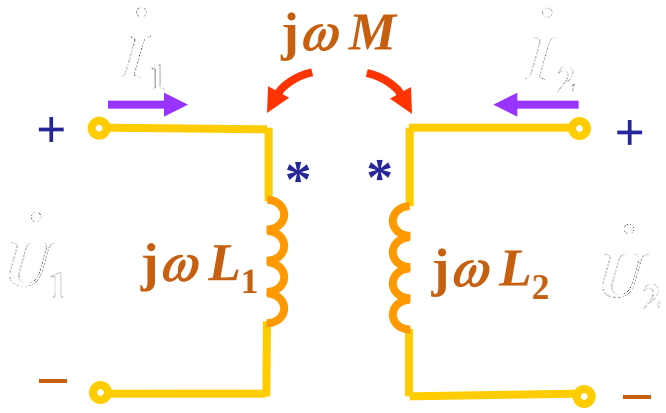


理想变压器模型

$$\begin{cases} u_1 = nu_2 \\ i_1 = -\dfrac{1}{n} i_2 \end{cases}$$

$$\begin{cases} \dot{U}_1 = n\dot{U}_2 \\ \dot{I}_1 = -\dfrac{1}{n} \dot{I}_2 \end{cases}$$

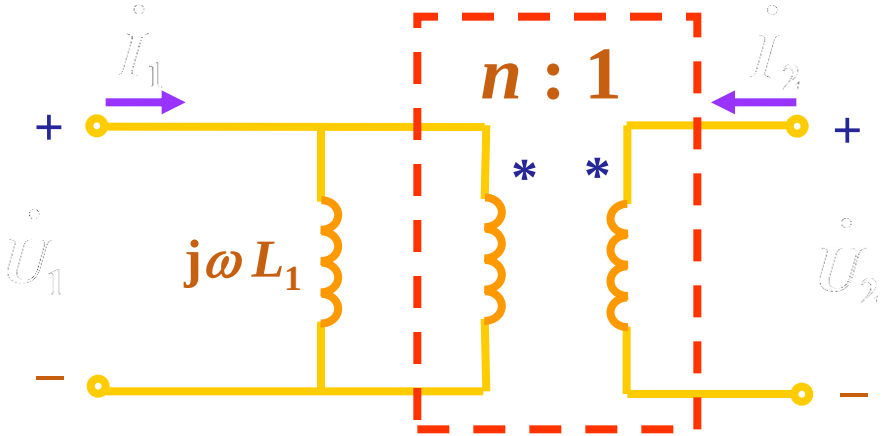
2.全耦合变压器(k=1，无损，L_1、L_2不为无限大)



由于全耦合，所以仍满足：

$$\frac{\dot{U}_1}{\dot{U}_2} = n = N_1 / N_2$$

又因 $\dot{U}_1 = \mathrm{j}\omega L_1 \dot{I}_1 + \mathrm{j}\omega M \dot{I}_2$



理想变压器

$$\dot{I}_1 = \frac{1}{\mathrm{j}\omega L_1}\dot{U}_1 - \frac{1}{n}\dot{I}_2$$

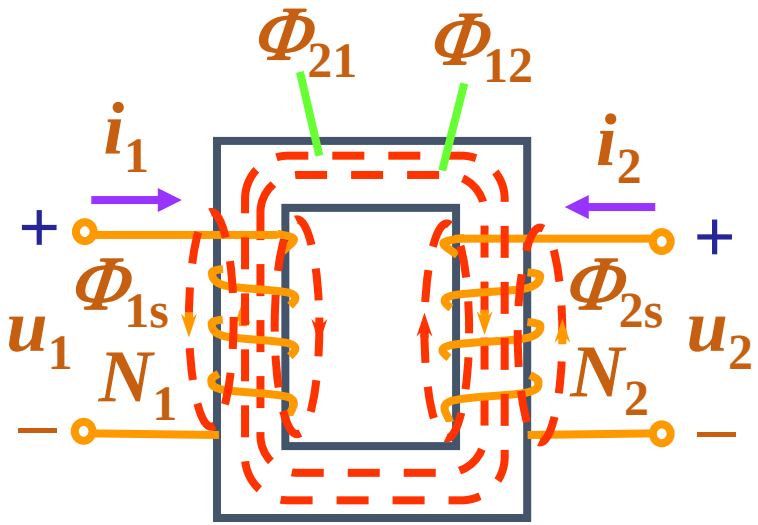
$\dot{I}_0$（空载激磁电流）

L_1：激磁电感

(*magnetizing inductance*)

全耦合变压器的等值电路图

3.无损非全耦合变压器(忽略损耗，$k\neq1$，$L\neq\infty$)



线圈中的磁通看成是漏磁通加全耦合磁通，即：

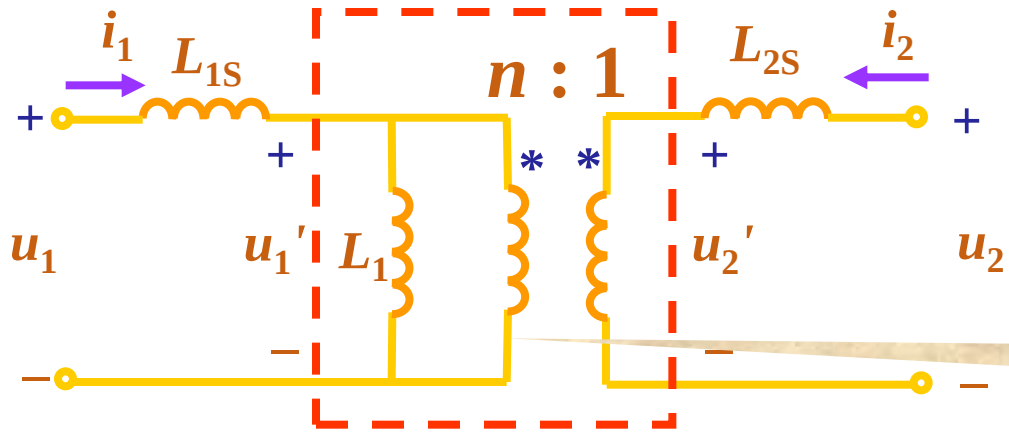
$$\begin{cases}\phi_1=\phi_{1S}+\phi\\ \phi_2=\phi_{2S}+\phi\end{cases}$$

全耦合磁通

在线性情况下，有：

$$\begin{cases} u_1=N_1\dfrac{\mathrm{d}\phi_1}{\mathrm{d}t}=\dfrac{\mathrm{d}\Psi_1}{\mathrm{d}t}=\dfrac{\mathrm{d}\Psi_{1S}}{\mathrm{d}t}+\dfrac{\mathrm{d}\Psi}{\mathrm{d}t}=L_{1S}\dfrac{di_1}{dt}+u_1' \\ u_2=N_2\dfrac{\mathrm{d}\phi_2}{\mathrm{d}t}=\dfrac{\mathrm{d}\Psi_2}{\mathrm{d}t}=\dfrac{\mathrm{d}\Psi_{2S}}{\mathrm{d}t}+\dfrac{\mathrm{d}\Psi}{\mathrm{d}t}=L_{2S}\dfrac{di_2}{dt}+u_2' \end{cases}$$

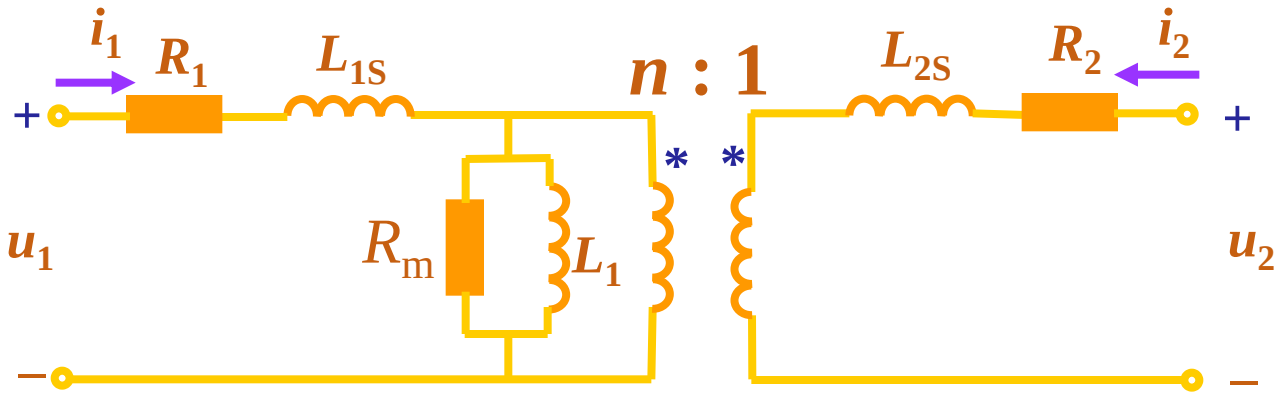
由此得无损非全耦合变压器的电路模型：



L_{1S}, L_{2S}：漏电感
(*leakage inductance*)

全耦合变压器

4. 有损耗的非全耦合变压器($k\neq1$，$L\neq\infty$， 线性)



考虑了导线和铁芯损耗